MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR,
ANALOG-TO-DIGITAL CONVERTERS WITH 3-STATE BUFFERED
10 BIT OUTPUTS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic 10-bit successive approximation A/D converters with 3-state buffered outputs. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit	Input voltage range
01	A/D Converter, 10 Bit	0 V to + 10 V unipolar -5 V to + 5 V bipolar

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
V	D-6 (18 lead, 1/4" x 15/16") dual-in-line package

1.3 Absolute maximum ratings.

Negative supply voltage (V_{EE}) to digital return- -	-16.5 V dc
Positive supply voltage (V_{CC}) to digital return- -	+7 V dc
Analog input voltage to analog return- - - - - -	±15 V dc
Digital input voltage to digital return- - - - - -	0 V to V_{CC}
Digital output voltage to digital return - - - - -	0 V to V_{CC}
Analog return to digital return- - - - - - - - - -	±1 V
Power dissipation- - - - - - - - - - - - - - - -	800 mW
Lead temperature (Soldering, 60 seconds) - - - - -	300°C
Storage temperature range- - - - - - - - - - - -	-65°C to +150°C
Junction temperature (T_J)- - - - - - - - - - - -	+175°C

1.4 Recommended operating conditions.

Positive supply voltage range- - - - - - - - - -	+4.5 V dc to +5.5 V dc
Negative supply voltage range- - - - - - - - - -	-13.5 V dc to 16.0 V dc
Case operating temperature range - - - - - - - - -	-55°C to +125°C
Minimum high-level input voltage - - - - - - - - -	2.0 V
Maximum low-level input voltage- - - - - - - - -	0.8 V

1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation	Maximum θJC	Maximum θJA
18 lead Dual-in-line	V	800 mW at T_A = 125°C	40°C/W	120°C/W

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Functional schematic circuit. The schematic circuit shall be as specified on Figure 2.

3.2.3 Block diagram. The block diagram shall be as specified on figure 3.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.2.5 Package and sealing material. Package and sealing material shall be in accordance with MIL-M-38510.

3.2.6 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The following electrical performance characteristics shall apply unless otherwise stated in table I:

V_{CC} = 5.0 V
V_{EE} = -15.0 V
Input logic "0" = 0.8 V max.
Input logic "1" = 2.0 V min.
$-55^{\circ}C \leq T_A \leq +125^{\circ}C$
Unipolar operation
A 15 Ω fixed resistor should be in series with DUT pin 13.

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3.5 Electrical test requirements. The electrical test requirements shall be as specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.2 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 57 (see MIL-M-38510, appendix E).

TABLE I. Electrical performance characteristics.

Test 1/	Symbol	Conditions (3.4 and fig. 7 - unless otherwise specified)	Sub group	Limits Min	Limits Max	Units
Conversion time	tc	V_{CC} = 5.0 V Bpc = 0 V (unipolar) Output code = any code	1,2,3 2/	15	40	µs
Power supply current from V_{CC}	I_{CC}	V_{CC} = 5.0 V 4/ V_{in} (analog) = 0 V Output code = 00 0000 0000	1,2,3	1	10	mA
Power supply current from V_{EE}	I_{EE}	V_{EE} = -15.0 V V_{in} (analog) = 0 V Output code = 00 0000 0000	1,2,3	-15	-1	"
Input low current	I_{IL}	V_{CC} = 5.5 V V_{in} (logic) = 0.0 V	1,2,3	-40	+1	µA
Input high current	I_{IH}	V_{CC} = 5.5 V V_{in} (logic) = 2.4 V	1,2,3	-1	40	"
		V_{in} (logic) = 5.5 V		-1	40	"
High impedence state output current	I_{ZH}	V_{CC} = 5.5 V Output bits 1 through 10 Vo = 2.4 V	1,2,3	-40	+40	"
	I_{ZL}	V_{CC} = 5.5 V Output bits 1 through 10 Vo = 0.4 V		-40	+40	"

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test 1/	Symbol	Conditions (3.4 and fig. 7 - unless otherwise specified)	Sub group	Limits Min	Limits Max	Units
B & $\overline{C}$ input logic threshold high	V_{IH}	B & $\overline{C}$ = 2.0 V Measure $\overline{DR}$ I_L = 0	1,2,3	2.4		V
B & $\overline{C}$ input logic threshold low	V_{IL}	B & $\overline{C}$ = 0.8 V Measure $\overline{DR}$ I_L = 0	1,2,3		0.4	"
Output logic voltage levels	V_{OH}	V_{CC} = 4.5 V All outputs except $\overline{DR}$ I_L = -.5 mA	1,2,3	2.4	-	"
		$\overline{DR}$ output I_L = -.25 mA		2.4	-	"
	V_{OL}	V_{CC} = 4.5 V I_L = 3.2 mA	1,2,3	-	0.4	"
Zero error at V_{in} = 0 V	Z_E	V_{CC} = 5 V ±.05 V V_{EE} = -15 V ± .015 V BpC = 0 V (unipolar) Output transition = 00 0000 000ø -55°C ≤ T_A ≤ 125°C T_A = 25°C End point See table IV 5/	4,5,6	 -3 -1 -2.5	 3 1 2.5	 LSB " "
	B_{ZE}	BpC = Open (bipolar) Output transition = øø øøøø øøøø -55°C ≤ T_A ≤ 125°C T_A = 25°C End point See table IV 5/ (Figures 9,10,11)	4,5,6	 -3 -1 -3.0	 3 1 3.0	 LSB " "
End-point accuracy	$-V_{FSE}$	V_{CC} = 5 V ±.05 V V_{EE} = -15 V ±.015 V BpC = Open (bipolar) Output transition = 00 0000 000ø -55°C ≤ T_A ≤ 125°C T_A = 25°C End point See table IV 5/	4,5,6	 -8.7 -3.7 -8.2	 8.7 3.7 8.2	 LSB " "
	$+V_{FSE}$	V_{CC} = 5 V ±.05 V V_{EE} = -15 V ±.015 V BpC = 0 V (unipolar) Output transition = 11 1111 111ø -55°C ≤ T_A ≤ 125°C T_A = 25°C End point See table IV 5/	4,5,6	 -8.7 -3.7 -8.2	 8.7 3.7 8.2	 LSB " "

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test 1/	Symbol	Conditions (3.4 and fig. 7 - unless otherwise specified)	Sub group	Limits Min	Limits Max	Units
End-point accuracy	$+V_{FSE}$	BpC = open (bipolar) Output transition = 11 1111 1111 -55°C ≤ T_A ≤ 125°C T_A = 25°C End point See table IV 5/ (Figures 9, 10, 11)	4,5,6	-8.7 -3.7 -8.2	8.7 3.7 8.2	LSB " "
Integral transition linearity error	T_{LE}	V_{CC} = 5 V ±.05 V V_{EE} = -15 V ±.015 V (Abbreviated test) BpC = 0 V (unipolar) -55°C ≤ T_A ≤ 125°C T_A = 25°C (Figures 9, 10, 11)	4,5,6	-0.5 "	0.5 "	" "
Differential linearity error	D_{LE}	V_{CC} = 5 V ±.05 V V_{EE} = -15 V ±.015 V BpC = 0 V (unipolar) -55°C ≤ T_A ≤ 125°C T_A = 25°C (Figures 9, 10, 11)	4,5,6	-0.9 "	1.0 "	" "
Power supply sensitivity to V_{EE}	$-P_{SS}$	V_{EE} = -16.00 V, -13.50 V BpC = 0 V (unipolar) Output transition = 00 0000 0000 -55°C ≤ T_A ≤ 125°C T_A = 25°C 3/	4,5,6	-0.18 -0.12	0.18 0.12	LSB/%PS "
		BpC = 0 V (unipolar) Output transition = 11 1111 1111 -55°C ≤ T_A ≤ 125°C T_A = 25°C (Figures 9, 10, 11)		-0.18 -0.12	0.18 0.12	" "
Power supply sensitivity to V_{CC}	$+P_{SS}$	V_{CC} = 4.5 V, 5.5 V BpC = 0 V (unipolar) Output transition = 00 0000 0000 -55°C ≤ T_A ≤ 125°C T_A = 25°C	4,5,6	-0.15 -0.1	0.15 0.1	" "
		BpC = 0 V (unipolar) Output transition = 11 1111 1111 -55°C ≤ T_A ≤ 125°C T_A = 25°C (Figures 9, 10, 11)		-0.15 -0.1	0.15 0.1	" "

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test 1/	Symbol	Conditions (3.4 and fig. 7 - unless otherwise specified)	Sub group	Limits Min	Limits Max	Units
Integral transition linearity error	T_{LE}	V_{CC} = 5 V ±.05 V V_{EE} = -15 V ±.015 V (All-codes test) BpC = 0 V (unipolar) -55°C ≤ T_A ≤ 125°C T_A = 25°C (Figures 9, 10, 11)	7, 8	-0.5	0.5	LSB
Data output delay	t_{DO}	Bit 1 through bit 10 outputs Output code = 00 0000 0000 Output code = 11 1111 1111 (Figures 4, 8, 9)	9	-	500	ns
Data blanking delay	t_{bK}	Bit 1 through bit 10 outputs Measured from $\overline{DR}$ low to high Output code = 00 0000 0000 Output code = 11 1111 1111 (Figures 4, 8, 9)	9	0.1	10.0	µs
Data ready delay	$\overline{t_{DR}}$	$\overline{DR}$ low to high measured from B & $\overline{C}$ low to high	9	0.1	1.5	"
Input resistance	Ri	Vin(analog) = 1/2 VFS BpC = open (bipolar) T_A = 25°C B & C = "0"	9	3	7	kohms

1/ See paragraph 6.5 for symbols and definitions.

2/ Unless otherwise state, the test measurements made in subgroups 1, 2, and 3 shall be made with the blank and convert input signal held low.

3/ ø represents the transition point between two adjacent code-words (i.e.: 00 0000 000ø represents the transition between code-words 00 0000 0000 and 00 0000 0001, 00 00øø øøøø represents the transition between code-words 00 0001 1111 and 00 0010 0000 and øø øøøø øøøø represents the transition.

4/ The initial turn on current may momentarily be in excess of 150 mA, if V_{CC} is applied before V_{EE}. Due to the nature of monolithic junction isolation; the application of V_{EE} prior or in conjuction with V_{CC} is preferred.

5/ The end point limits of table IV shall be used for design.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

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4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in tests (method 1015 of MIL-STD-883).

(1) Class S devices: Static tests (test condition D) using the circuit shown on figure 6. Test duration shall be 240 hours minimum.

(2) Class B devices: Test condition D, using the circuit shown on figure 6.

b. Interim and final electrical test parameters shall be as specified in table II herein.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

TABLE II. Electrical test requirements.

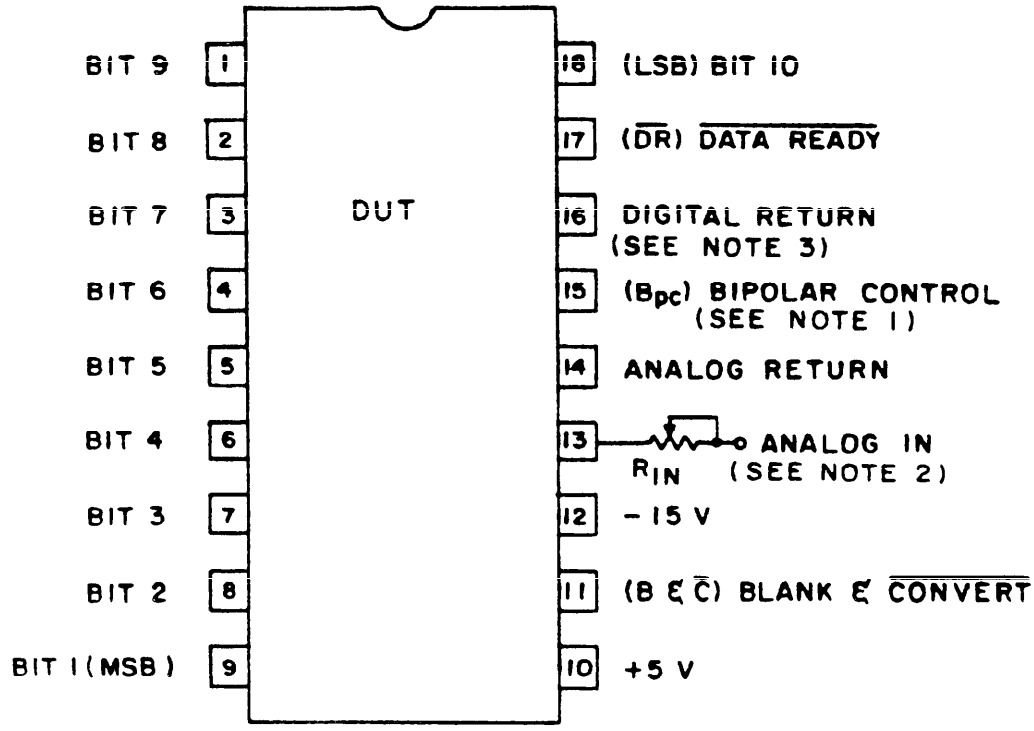
MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S 1/ 2/	Class B 1/ 2/
Interim electrical parameters (pre-burn-in) method 5004	1	1
Final electrical test parameters (method 5004)	1*, 2, 3, 4, 5, 6	1*, 2, 3, 4, 5, 6
Group A test requirements (method 5005)	1, 2, 3, 4, 5, 6, 7, 8, 9	1, 2, 3, 4, 5, 6, 9
Group C end-point and group B, class S, electrical parameters (method 5005)	1, 2, 3, 4, 5, 6 and table IV limits**	1, 4 and table IV limits**
Additional electrical subgroups for group C periodic inspection	N/A	7, 8
Group D end-point electrical parameters (method 5005)	1, 2, 3, 4, 5, 6 and table IV end point limits**	1, 4 and table IV end point limits**

1/ (*) indicates PDA applies to subgroup 1 (see 4.2c).

2/ See paragraph 4.4.4.

(**) Table IV end point limits shall apply where applicable.

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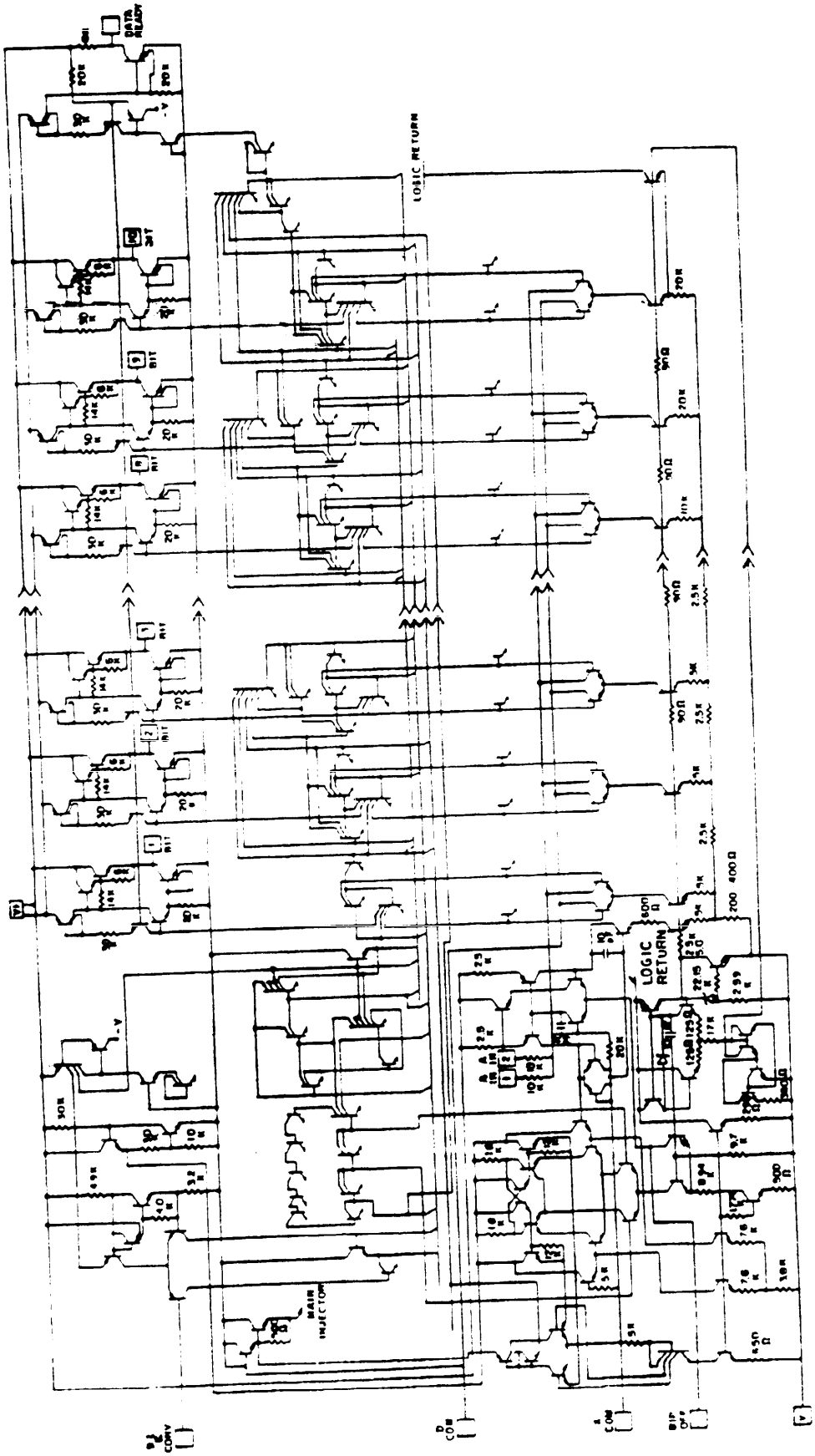
NOTES: 1. For unipolar operation, pin 15 shall be connected to pin 16; for bipolar operation, pin 15 shall be open.

2. R_{in} is a 15 ohm fixed or a 50 ohm variable resistor.

3. If the digital return and the analog return terminals are remotely connected, a parallel pair of back-to-back diodes shall be connected between pins 14 and 16.

FIGURE 1. Terminal connections.

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NOTE: Transistors drawn with an open lead (i.e. ( ) are connected to Logic Return.

FIGURE 2. Functional schematic circuits.

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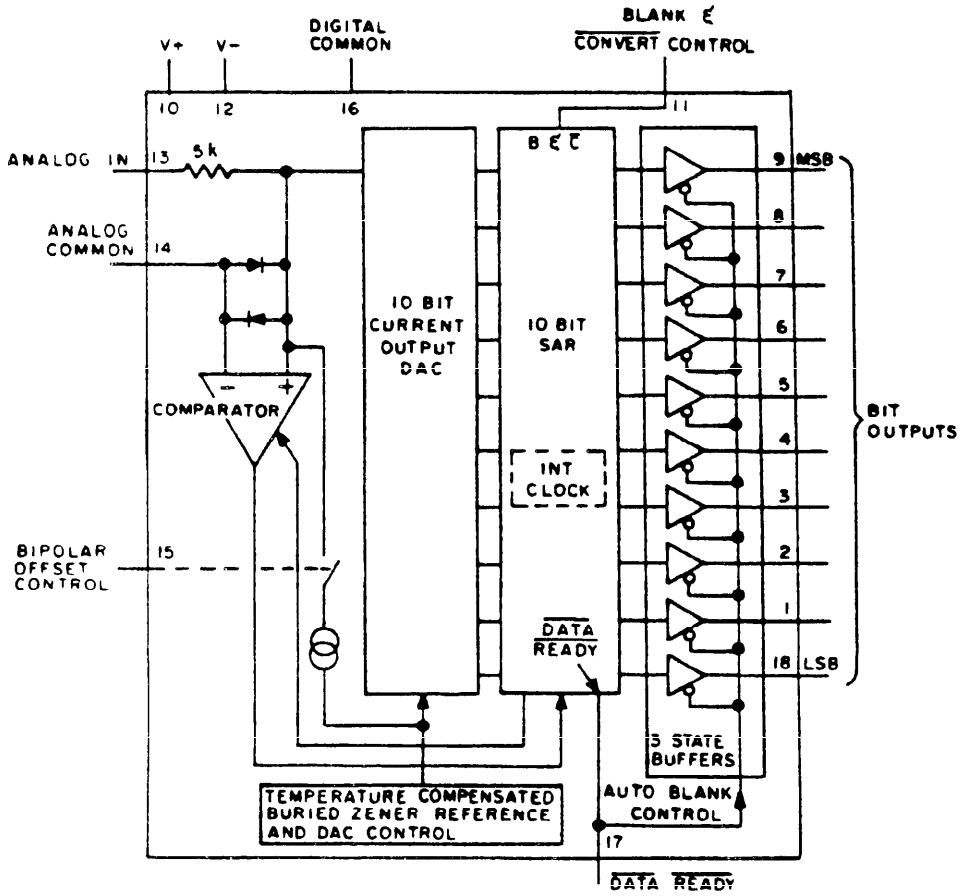
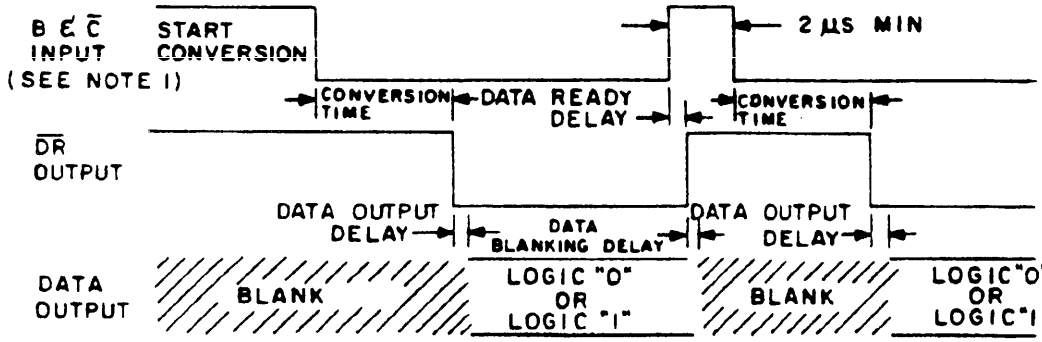


FIGURE 3. Block diagram.



TIMING & CONTROL SEQUENCE

NOTES: 1. The B and C start conversion pulse width shall be 2 µs minimum.

2. Data ready delay is 1.5 µs maximum.

FIGURE 4. Timing diagram.

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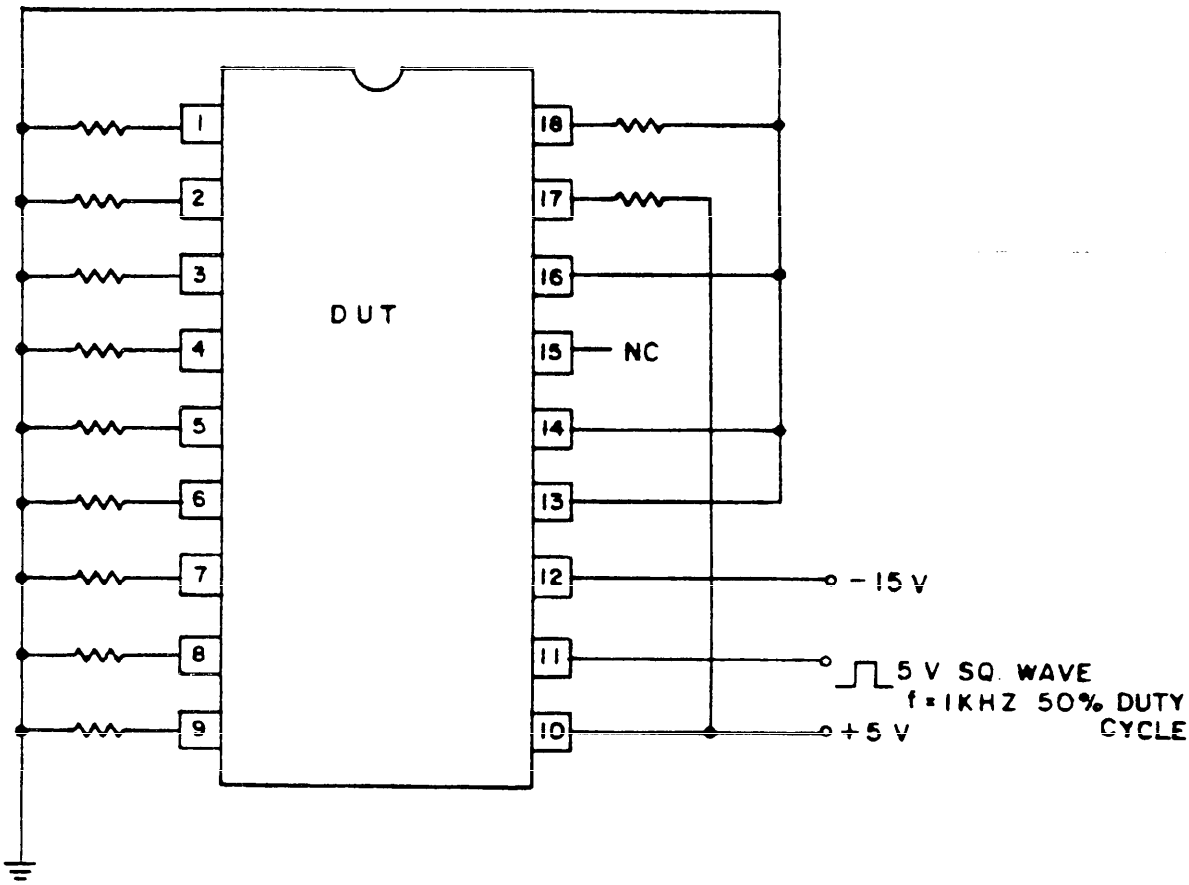
Analog Input		Digital Output	
Unipolar	Bipolar	MSB LSB 1/	
0.0098	-4.9951	00 0000 000Ø	2/
4.9902	-0.0146	01 1111 111Ø	
5.0000	-0.0049	ØØ ØØØØ ØØØØ	
5.0098	+0.0049	10 0000 000Ø	
9.9902	+4.9854	11 1111 111Ø	

1/ 1 LSB - 9.766 mV

2/ Ø represents the transition point between two adjacent code-words (ie: 00 0000 0000 represents the transition between code-words 00 0000 0000 and 00 0000 0001, and ØØ ØØØØ ØØØØ represents the transition between code words 01 1111 1111 and 10 0000 0000).

FIGURE 5. Digital output codes.

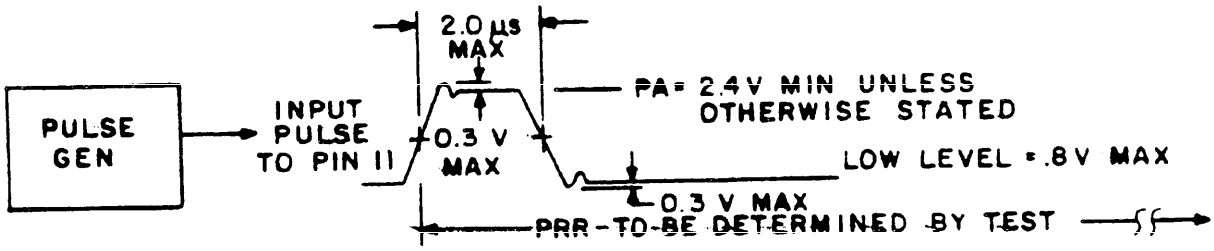
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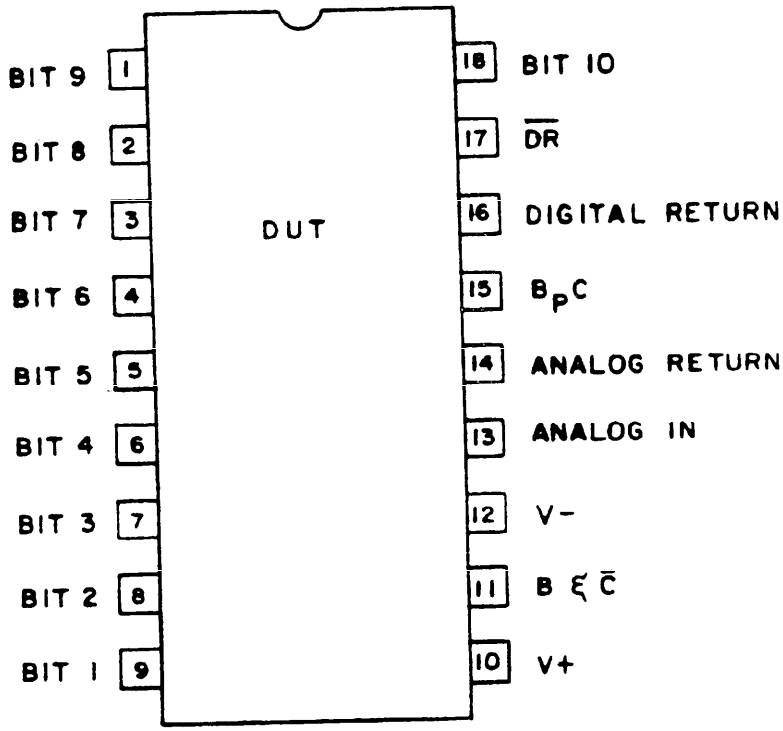


NOTES:
1. All resistors 3.0 K 1% 1/4W MF.
2. I_{CC} (+5 V) ∿ 1 Amp/100 devices.
   I_{EE} (-15 V) ∿ 2 Amp/100 devices.
3. 10 µf caps - one per supply.

FIGURE 6. Test circuit, burn-in and steady state life test.

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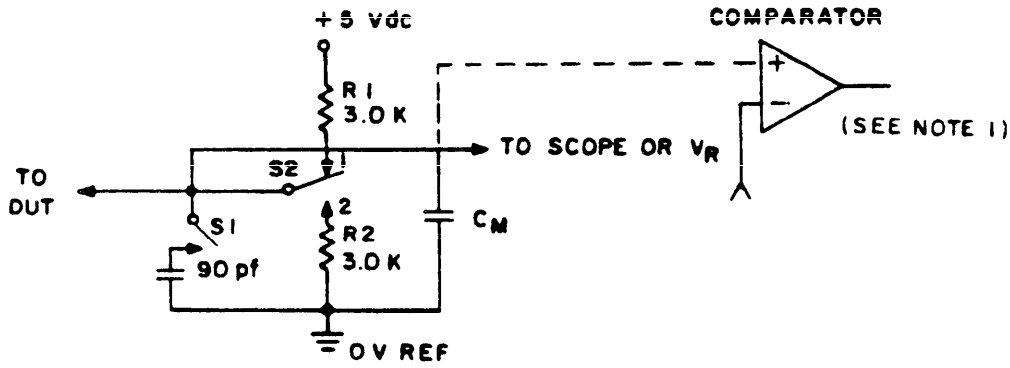




NOTES:
1. Connect pins 14 and 16 together to form a unipoint ground.
2. All voltages are referenced to 0 V.

FIGURE 7. Test circuit for static and dynamic tests.

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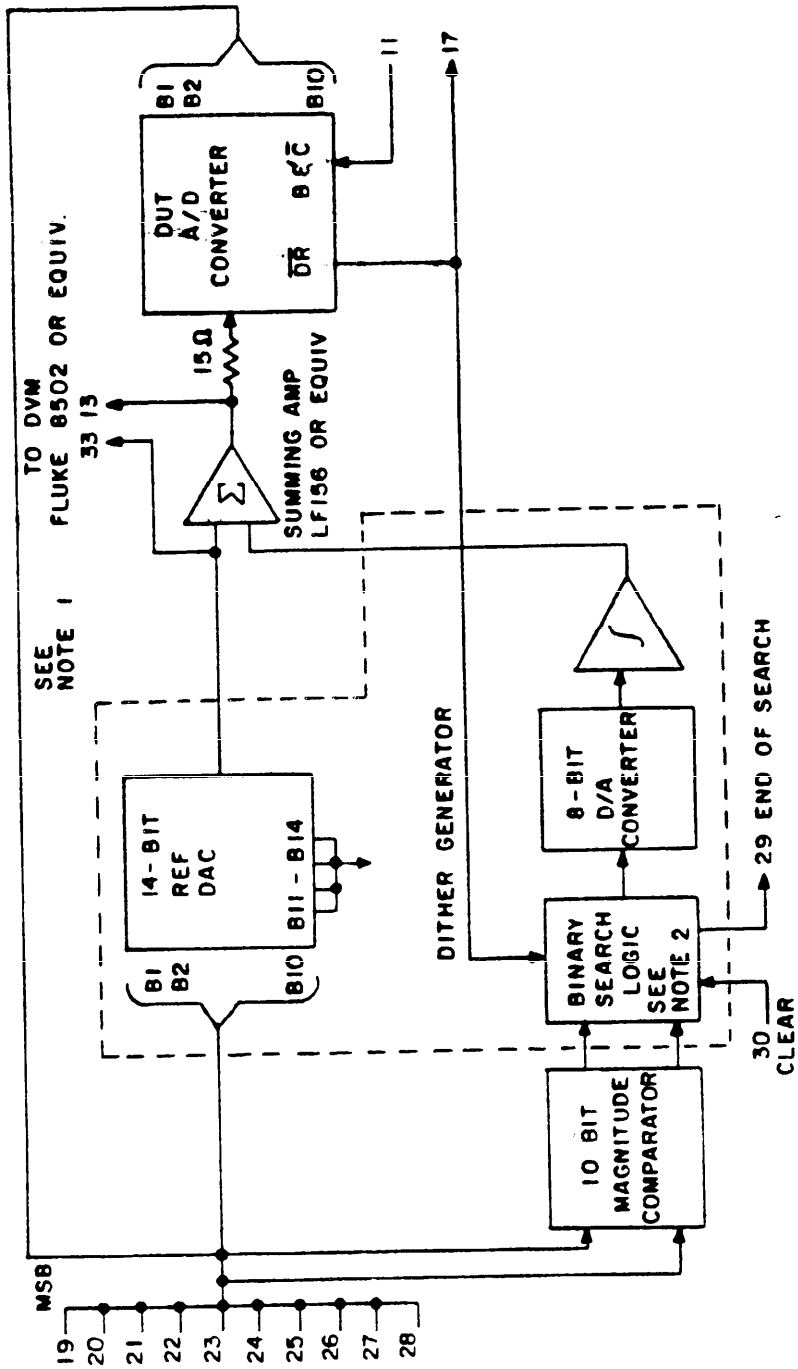
Logic load test circuit for all logic outputs except $\overline{DR}$. $\overline{DR}$ is an open collector NPN transistor with a 6 kΩ nominal pullup resistor. Test fixture loading should be greater than 20 kΩ and 15 pF or less.

NOTES:
1. Used in performing automatic test.
2. C_M = 10 ±2 pF (including probe and parasitic capacitance).
3. Load circuit shall be connected to each DUT digital output pin prior to performing t_{DO}.
4. For tests t_{DO} use the following table:

Transition	S1	S2		VR
3 → L	Closed	Pos 1	(WRT $\overline{DR}$ high to low)	0.4 V
3 → H	Closed	Pos 2	(WRT $\overline{DR}$ high to low)	2.4 V
L → 3	Open	Pos 1	(WRT B and $\overline{C}$ low to high)	2.4 V
H → 3	Open	Pos 2	(WRT B and $\overline{C}$ low to high)	0.4 V

FIGURE 8. Logic load test circuit.

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NOTES:
1. The circuit contained within the dashed outline is for fast automatic testing and is optional. The 8-bit D/A converter may be replaced with a bipolar current source that is controlled directly by the magnitude comparator.
2. A more detailed drawing of the binary search logic is presented in figure 11.
3. Adapter pins 1 through 18 correspond to DUT pins 1 through 18.
4. See figure 8 for blank and convert pulse waveform.

FIGURE 9. Block diagram of A/D converter test circuit.

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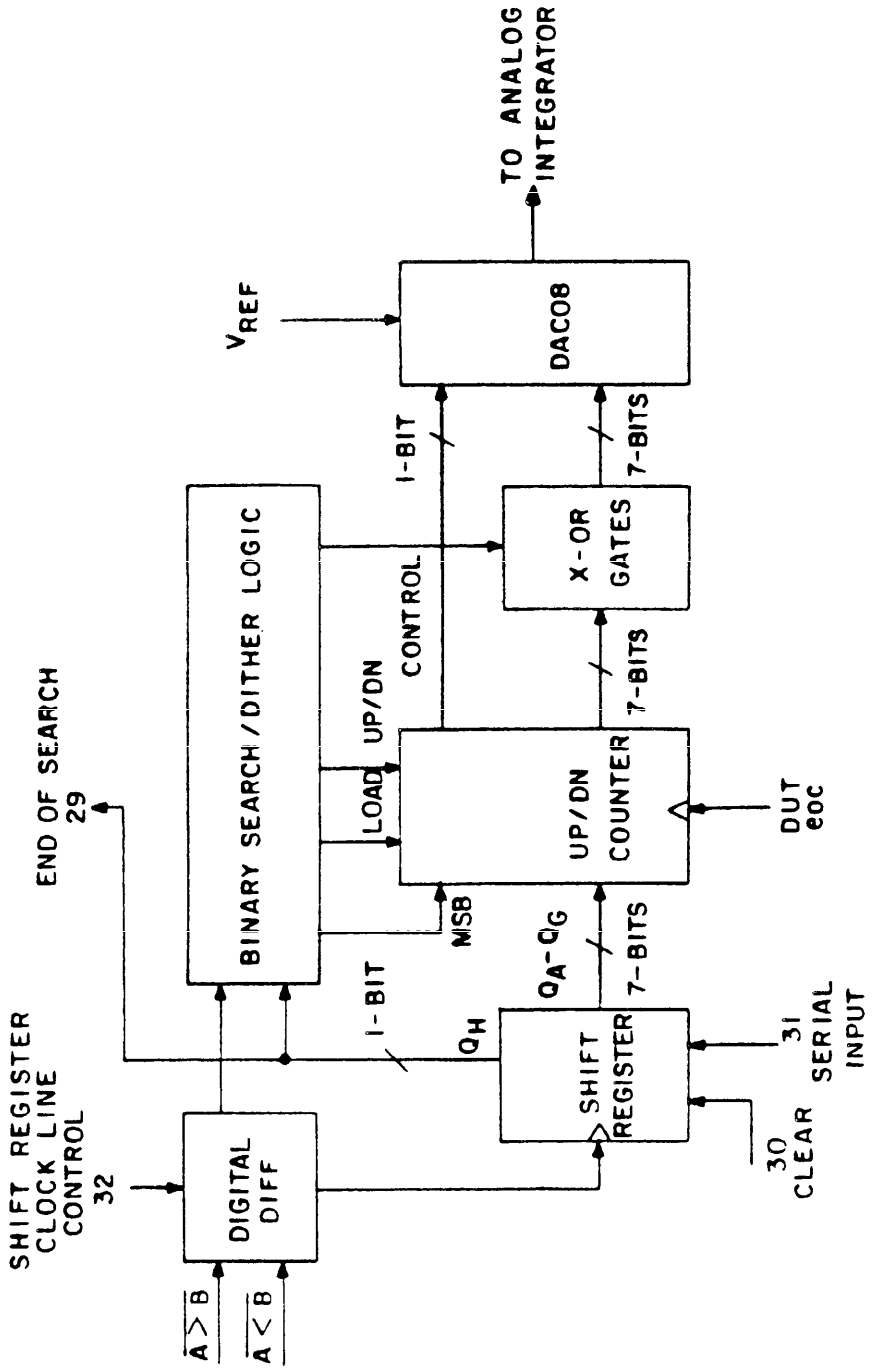
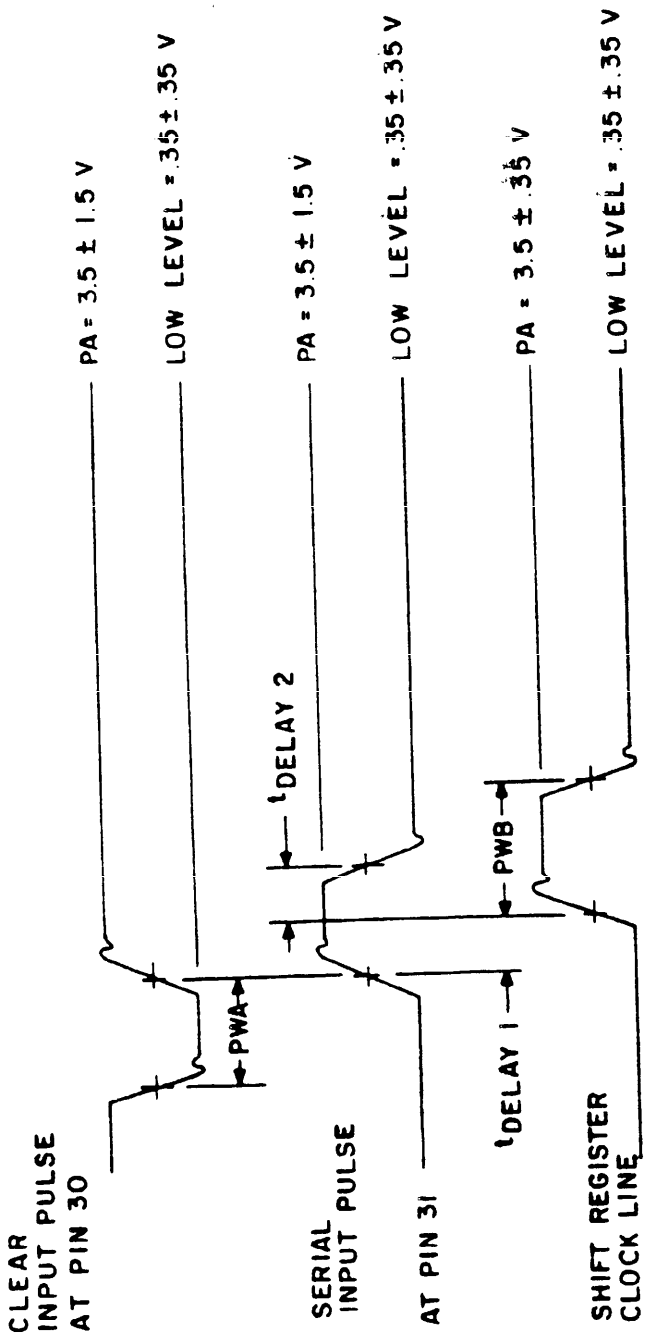


FIGURE 10. Block diagram of binary search and dither circuit.

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NOTES: 1. PWA = 2 µS(min)
PWB = 200 ns (min)
$t_{DELAY\ 1}$ = 500 ns (min
$t_{DELAY\ 2}$ = 500 ns (min)
2. The timing for the serial input pulse and the clock line input pulse should be established at the shift register (figure 11).

FIGURE 11. Reset waveforms for binary search and dither test circuit.

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TABLE III. Group A inspection.

Subgroup	Symbol	Test no.	Applied voltages and loads ref. pin no. 14 (figure 7)						Measurement sense lines ref. pin no. 14			Equations	Limits		Units
			I0 V_{CC} (V)	I2 V_{EE} (V)	I5 V_{IH} (V)	I3 $V_{in(a)}$ (V)	Output pin	Output load	Pin	Value	Unit		Min	Max	
T_A = 25°C	t_c	1	+5.00	-15.00		MAX(-)	-		17	t_1	µs	$t_c = t_1$	15	40	µs
	I_{CC1}	2	"	"	"	"	"	"	10	I_1	mA	$I_{CC1} = I_1$	1	10	mA
	I_{EE1}	3	"	"	"	"	"	"	12	I_2	"	$I_{EE1} = I_2$	-25	1	"
	I_{IL}	4	+5.50	"	0.0	"	"	"	11	I_3	"	$I_{IL} = I_3$	-40	+1	µA
	I_{IH1}	5	"	"	2.4	"	"	"	"	I_4	µA	$I_{IH1} = I_4$	-1	40	"
	I_{IH2}	6	"	"	5.5	"	"	"	"	I_5	"	$I_{IH2} = I_5$	"	"	"
	I_{ZH1}	7	"	"	1	MAX(-)	9	V_o = 2.4 V	9	I_6	"	$I_{ZH1} = I_6$	-40	"	"
	I_{ZH2}	8	"	"	"	"	8	"	8	I_7	"	$I_{ZH2} = I_7$	"	"	"
	I_{ZH3}	9	"	"	"	"	7	"	7	I_8	"	$I_{ZH3} = I_8$	"	"	"
	I_{ZH4}	10	"	"	"	"	6	"	6	I_9	"	$I_{ZH4} = I_9$	"	"	"
	I_{ZH5}	11	"	"	"	"	5	"	5	I_{10}	"	$I_{ZH5} = I_{10}$	"	"	"
	I_{ZH6}	12	"	"	"	"	4	"	4	I_{11}	"	$I_{ZH6} = I_{11}$	"	"	"
	I_{ZH7}	13	"	"	"	"	3	"	3	I_{12}	"	$I_{ZH7} = I_{12}$	"	"	"
	I_{ZH8}	14	"	"	"	"	2	"	2	I_{13}	"	$I_{ZH8} = I_{13}$	"	"	"
	I_{ZH9}	15	"	"	"	"	1	"	1	I_{14}	"	$I_{ZH9} = I_{14}$	"	"	"
	I_{ZH10}	16	"	"	"	"	18	"	18	I_{15}	"	$I_{ZH10} = I_{15}$	"	"	"
	I_{ZL1}	17	"	"	"	"	9	V_o = 0.4 V	9	I_{16}	"	$I_{ZL1} = I_{16}$	"	"	"
	I_{ZL2}	18	"	"	"	"	8	"	8	I_{17}	"	$I_{ZL2} = I_{17}$	"	"	"
	I_{ZL3}	19	"	"	"	"	7	"	7	I_{18}	"	$I_{ZL3} = I_{18}$	"	"	"
	I_{ZL4}	20	"	"	"	"	6	"	6	I_{19}	"	$I_{ZL4} = I_{19}$	"	"	"
	I_{ZL5}	21	"	"	"	"	5	"	5	I_{20}	"	$I_{ZL5} = I_{20}$	"	"	"
	I_{ZL6}	22	"	"	"	"	4	"	4	I_{21}	"	$I_{ZL6} = I_{21}$	"	"	"
	I_{ZL7}	23	"	"	"	"	3	"	3	I_{22}	"	$I_{ZL7} = I_{22}$	"	"	"
	I_{ZL8}	24	"	"	"	"	2	"	2	I_{23}	"	$I_{ZL8} = I_{23}$	"	"	"
	I_{ZL9}	25	"	"	"	"	1	"	1	I_{24}	"	$I_{ZL9} = I_{24}$	"	"	"
	I_{ZL10}	26	"	"	"	"	18	"	18	I_{25}	"	$I_{ZL10} = I_{25}$	"	"	"
	V_{IH}	27	+5.0	"	2.0		17	"	17	E_1	V	$V_{IH} = E_1$	2.4	-	V
	V_{IL}	28	+5.0	"	0.8		17	0	17	E_2	V	$V_{IL} = E_2$	-	0.4	"

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TABLE III. Group A Inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltages and loads ref. pin no. 14 (Figure 7)						Measurement sense lines ref. pin no. 14			Equations	Limits		Units
			I5 V_{CC} (V)	I2 V_{EE} (V)	I5 A & C (V)	I3 Vin(a) (V)	Output Pin	Output Load	Pin	Value	Unit		Min	Max	
1 T_A = 25°C	V_{OH1}	29	+4.50	-15.00	-	MAX(+)	9	-.5 mA	9	E_3	V	$V_{OH1} = E_3$	2.4	-	V
	V_{OH2}	30	"	"	"	"	8	"	8	E_4	"	$V_{OH2} = E_4$	"	"	"
	V_{OH3}	31	"	"	"	"	7	"	7	E_5	"	$V_{OH3} = E_5$	"	"	"
	V_{OH4}	32	"	"	"	"	6	"	6	E_6	"	$V_{OH4} = E_6$	"	"	"
	V_{OH5}	33	"	"	"	"	5	"	5	E_7	"	$V_{OH5} = E_7$	"	"	"
	V_{OH6}	34	"	"	"	"	4	"	4	E_8	"	$V_{OH6} = E_8$	"	"	"
	V_{OH7}	35	"	"	"	"	3	"	3	E_9	"	$V_{OH7} = E_9$	"	"	"
	V_{OH8}	36	"	"	"	"	2	"	2	E_{10}	"	$V_{OH8} = E_{10}$	"	"	"
	V_{OH9}	37	"	"	"	"	1	"	1	E_{11}	"	$V_{OH9} = E_{11}$	"	"	"
	V_{OH10}	38	"	"	"	"	18	"	18	E_{12}	"	$V_{OH10} = E_{12}$	"	"	"
	V_{OH11}	39	"	"	"	"	17	-.25 mA	17	E_{13}	"	$V_{OH11} = E_{13}$	"	"	"
	V_{OL1}	40	"	"	"	MAX(-)	9	3.2 mA	9	E_{14}	"	$V_{OL1} = E_{14}$	-	0.4	"
	V_{OL2}	41	"	"	"	"	8	"	8	E_{15}	"	$V_{OL2} = E_{15}$	"	"	"
	V_{OL3}	42	"	"	"	"	7	"	7	E_{16}	"	$V_{OL3} = E_{16}$	"	"	"
	V_{OL4}	43	"	"	"	"	6	"	6	E_{17}	"	$V_{OL4} = E_{17}$	"	"	"
	V_{OL5}	44	"	"	"	"	5	"	5	E_{18}	"	$V_{OL5} = E_{18}$	"	"	"
	V_{OL6}	45	"	"	"	"	4	"	4	E_{19}	"	$V_{OL6} = E_{19}$	"	"	"
	V_{OL7}	46	"	"	"	"	3	"	3	E_{20}	"	$V_{OL7} = E_{20}$	"	"	"
	V_{OL8}	47	"	"	"	"	2	"	2	E_{21}	"	$V_{OL8} = E_{21}$	"	"	"
	V_{OL9}	48	"	"	"	"	1	"	1	E_{22}	"	$V_{OL9} = E_{22}$	"	"	"
	V_{OL10}	49	"	"	"	"	18	"	18	E_{23}	"	$V_{OL10} = E_{23}$	"	"	"
	V_{OL11}	50	"	"	"	"	17	"	17	E_{24}	"	$V_{OL11} = E_{24}$	"	"	"
2 T_A = 125°C	(All test parameters, test conditions, equations, and test limits are identical with those specified in table III, subgroup 1.)														
3 T_A = -55°C	(All test parameters, test conditions, equations, and test limits are identical with those specified in table III, subgroup 1.)														

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TABLE III. Group A Inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltages ref. pin no. 14 (Figures 8, 9) 10 V_{CC} (V)	12 V_{EE} (V)	15 BpC (V)	Applied digital code-word 19<------>28 MSB<------>LSB	Output digital code-word 9<---->1,19,17 MSB<------>LSB	Measurement sense lines ref. pin no. 14 Pin	Value	Unit	Equations	Limits Min	Max	Units
4 T_A = 25°C	Z_E	151	+5.00	-15.00	0	00 0000 0000	00 0000 0000	13	E_{73}	V	Z_E = 102.4*(E_{73})	-1	+1	LSB
	B_{7E}	152	"	"	Open	01 1111 1111	dd dddd dddd	"	E_{74}	"	Z_E = 102.4*(E_{74})	"	"	"
	$-V_{FSE}$	153	"	"	Open	00 0000 0000	00 0000 000d	"	E_{75}	"	$-V_{FSE}$ = 102.4*(E_{75} +5 +10/1024)	-3.7	+3.7	"
	$+V_{FSE}$	154	"	"	0	11 1111 1110	11 1111 111d	"	E_{76}	"	$+V_{FSE}$ = 102.4*(E_{76} -10 -10/1024)	"	"	"
		155	"	"	"	" " "	" " "	"	E_{77}	"	$+V_{FSE}$ = 102.4*(E_{77} -5 -10/1024)	"	"	"
											For end-point linearity error, TLE(N)=(E(N+79)-V(ideal))/S where S = (E_{79} - E_{78})/1022 for N = 1 to 98			
			"	"	"	00 0000 0000	00 0000 000d	"	E_{78}	"				
			"	"	"	11 1111 1110	11 1111 111d	"	E_{79}	"				
	T_{LE1}	156	"	"	"	00 0000 0001	00 0000 00dd	"	E_{80}	"	T_{LE1} = (E_{80} - 1* S - E_{78})/S	-.50	+.50	"
	T_{LE2}	157	"	"	"	00 0000 0010	00 0000 001d	"	E_{81}	"	T_{LE2} = (E_{81} - 2* S - E_{78})/S	"	"	"
	T_{LE3}	158	"	"	"	00 0000 0011	00 0000 0ddd	"	E_{82}	"	T_{LE3} = (E_{82} - 3* S - E_{78})/S	"	"	"
	T_{LE4}	159	"	"	"	00 0000 0100	00 0000 010d	"	E_{83}	"	T_{LE4} = (E_{83} - 4* S - E_{78})/S	"	"	"
	T_{LE5}	160	"	"	"	00 0000 0101	00 0000 01dd	"	E_{84}	"	T_{LE5} = (E_{84} - 5* S - E_{78})/S	"	"	"
	T_{LE6}	161	"	"	"	00 0000 0110	00 0000 011d	"	E_{85}	"	T_{LE6} = (E_{85} - 6* S - E_{78})/S	"	"	"
	T_{LE7}	162	"	"	"	00 0000 0111	00 0000 dddd	"	E_{86}	"	T_{LE7} = (E_{86} - 7* S - E_{78})/S	"	"	"
	T_{LE8}	163	"	"	"	00 0000 1000	00 0000 100d	"	E_{87}	"	T_{LE8} = (E_{87} - 8* S - E_{78})/S	"	"	"
	T_{LE9}	164	"	"	"	00 0000 1001	00 0000 10dd	"	E_{88}	"	T_{LE9} = (E_{88} - 9* S - E_{78})/S	"	"	"
	T_{LE10}	165	"	"	"	00 0000 1010	00 0000 101d	"	E_{89}	"	T_{LE10} = (E_{89} - 10* S - E_{78})/S	"	"	"
	T_{LE11}	166	"	"	"	00 0000 1011	00 0000 1ddd	"	E_{90}	"	T_{LE11} = (E_{90} - 11* S - E_{78})/S	"	"	"
	T_{LE12}	167	"	"	"	00 0000 1101	00 0000 11dd	"	E_{91}	"	T_{LE12} = (E_{91} - 13* S - E_{78})/S	"	"	"
	T_{LE13}	168	"	"	"	00 0000 1110	00 0000 111d	"	E_{92}	"	T_{LE13} = (E_{92} - 14* S - E_{78})/S	"	"	"
	T_{LE14}	169	"	"	"	00 0000 1111	00 0000 dddd	"	E_{93}	"	T_{LE14} = (E_{93} - 15* S - E_{78})/S	"	"	"
	T_{LE15}	170	"	"	"	00 0001 0000	00 0001 000d	"	E_{94}	"	T_{LE15} = (E_{94} - 16* S - E_{78})/S	"	"	"
	T_{LE16}	171	"	"	"	00 0001 0001	00 0001 00dd	"	E_{95}	"	T_{LE16} = (E_{95} - 17* S - E_{78})/S	"	"	"
	T_{LE17}	172	"	"	"	00 0001 0010	00 0001 001d	"	E_{96}	"	T_{LE17} = (E_{96} - 18* S - E_{78})/S	"	"	"
	T_{LE18}	173	"	"	"	00 0001 0011	00 0001 0ddd	"	E_{97}	"	T_{LE18} = (E_{97} - 19* S - E_{78})/S	"	"	"
	T_{LE19}	174	"	"	"	00 0001 0101	00 0001 11dd	"	E_{98}	"	T_{LE19} = (E_{98} - 29* S - E_{78})/S	"	"	"
	T_{LE20}	175	"	"	"	00 0001 1110	00 0001 111d	"	E_{99}	"	T_{LE20} = (E_{99} - 30* S - E_{78})/S	"	"	"
	T_{LE21}	176	"	"	"	00 0001 1111	00 00dd dddd	"	E_{100}	"	T_{LE21} = (E_{100} - 31* S - E_{78})/S	"	"	"
	T_{LE22}	177	"	"	"	00 0010 0000	00 0010 000d	"	E_{101}	"	T_{LE22} = (E_{101} - 32* S - E_{78})/S	"	"	"
	T_{LE23}	178	"	"	"	00 0010 0001	00 0010 00dd	"	E_{102}	"	T_{LE23} = (E_{102} - 33* S - E_{78})/S	"	"	"
	T_{LE24}	179	"	"	"	00 0010 0010	00 0010 001d	"	E_{103}	"	T_{LE24} = (E_{103} - 34* S - E_{78})/S	"	"	"
	T_{LE25}	180	"	"	"	00 0010 0011	00 0010 0ddd	"	E_{104}	"	T_{LE25} = (E_{104} - 35* S - E_{78})/S	"	"	"
	T_{LE26}	181	"	"	"	00 0011 1101	00 0011 11dd	"	E_{105}	"	T_{LE26} = (E_{105} - 61* S - E_{78})/S	"	"	"
	T_{LE27}	182	"	"	"	00 0011 1110	00 0011 111d	"	E_{106}	"	T_{LE27} = (E_{106} - 62* S - E_{78})/S	"	"	"
	T_{LE28}	183	"	"	"	00 0011 1111	00 0ddd dddd	"	E_{107}	"	T_{LE28} = (E_{107} - 63* S - E_{78})/S	"	"	"
	T_{LE29}	184	"	"	"	00 0100 0000	00 0100 000d	"	E_{108}	"	T_{LE29} = (E_{108} - 64* S - E_{78})/S	"	"	"
	T_{LE30}	185	"	"	"	00 0100 0001	00 0100 00dd	"	E_{109}	"	T_{LE30} = (E_{109} - 65* S - E_{78})/S	"	"	"
	T_{LE31}	186	"	"	"	00 0100 0010	00 0100 001d	"	E_{110}	"	T_{LE31} = (E_{110} - 66* S - E_{78})/S	"	"	"
	T_{LE32}	187	"	"	"	00 0100 0011	00 0100 0ddd	"	E_{111}	"	T_{LE32} = (E_{111} - 67* S - E_{78})/S	"	"	"
	T_{LE33}	188	"	"	"	00 0111 1101	00 0111 11dd	"	E_{112}	"	T_{LE33} = (E_{112} - 125* S - E_{78})/S	"	"	"
	T_{LE34}	189	"	"	"	00 0111 1110	00 0111 111d	"	E_{113}	"	T_{LE34} = (E_{113} - 126* S - E_{78})/S	"	"	"
	T_{LE35}	190	"	"	"	00 0111 1111	00 dddd dddd	"	E_{114}	"	T_{LE35} = (E_{114} - 127* S - E_{78})/S	"	"	"

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TABLE III. Group A Inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltages ref. pin no. 14 (Figures 8, 9)			Applied digital code-word	Output digital code-word	Measurement sense lines ref. pin no. 14			Equations	Limits		Units
			19 V_{CC} (V)	12 V_{EE} (V)	15 BpC (V)	19<------>28 MSB<------>LSB	9<----->1,18,17 MSB<------>LSB	Pin	Value	Unit		Min	Max	
4 T_A = 25°C	T_{LE35}	191	+5.00	-15.00	0	00 1000 0000	00 1000 000ø	13	E_{115}	V	$T_{LE36} = (E_{115} - 128^* S - E_{78})/S$	-.50	+.50	LSB
	T_{LE37}	192	"	"	"	00 1000 0001	00 1000 00øø	"	E_{116}	"	$T_{LE37} = (E_{116} - 129^* S - E_{78})/S$	"	"	"
	T_{LE38}	193	"	"	"	00 1000 0010	00 1000 001ø	"	E_{117}	"	$T_{LE38} = (E_{117} - 130^* S - E_{78})/S$	"	"	"
	T_{LE39}	194	"	"	"	00 1000 0011	00 1000 0øøø	"	E_{118}	"	$T_{LE39} = (E_{118} - 131^* S - E_{78})/S$	"	"	"
	T_{LE40}	195	"	"	"	00 1111 1101	00 1111 11øø	"	E_{119}	"	$T_{LE40} = (E_{119} - 253^* S - E_{78})/S$	"	"	"
	T_{LE41}	196	"	"	"	00 1111 1110	00 1111 111ø	"	E_{120}	"	$T_{LE41} = (E_{120} - 254^* S - E_{78})/S$	"	"	"
	T_{LE42}	197	"	"	"	00 1111 1111	00 øøøø øøøø	"	E_{121}	"	$T_{LE42} = (E_{121} - 255^* S - E_{78})/S$	"	"	"
	T_{LE43}	198	"	"	"	01 0000 0000	01 0000 000ø	"	E_{122}	"	$T_{LE43} = (E_{122} - 256^* S - E_{78})/S$	"	"	"
	T_{LE44}	199	"	"	"	01 0000 0001	01 0000 00øø	"	E_{123}	"	$T_{LE44} = (E_{123} - 257^* S - E_{78})/S$	"	"	"
	T_{LE45}	200	"	"	"	01 0000 0010	01 0000 001ø	"	E_{124}	"	$T_{LE45} = (E_{124} - 258^* S - E_{78})/S$	"	"	"
	T_{LE46}	201	"	"	"	01 0000 0011	01 0000 0øøø	"	E_{125}	"	$T_{LE46} = (E_{125} - 259^* S - E_{78})/S$	"	"	"
	T_{LE47}	202	"	"	"	01 1111 1101	01 1111 11øø	"	E_{126}	"	$T_{LE47} = (E_{126} - 509^* S - E_{78})/S$	"	"	"
	T_{LE48}	203	"	"	"	01 1111 1110	01 1111 111ø	"	E_{127}	"	$T_{LE48} = (E_{127} - 510^* S - E_{78})/S$	"	"	"
	T_{LE49}	204	"	"	"	01 1111 1111	øø øøøø øøøø	"	E_{128}	"	$T_{LE49} = (E_{128} - 511^* S - E_{78})/S$	"	"	"
	T_{LE50}	205	"	"	"	10 0000 0000	10 0000 000ø	"	E_{129}	"	$T_{LE50} = (E_{129} - 512^* S - E_{78})/S$	"	"	"
	T_{LE51}	206	"	"	"	10 0000 0001	10 0000 00øø	"	E_{130}	"	$T_{LE51} = (E_{130} - 513^* S - E_{78})/S$	"	"	"
	T_{LE52}	207	"	"	"	10 0000 0010	10 0000 001ø	"	E_{131}	"	$T_{LE52} = (E_{131} - 514^* S - E_{78})/S$	"	"	"
	T_{LE53}	208	"	"	"	10 0000 0011	10 0000 0øøø	"	E_{132}	"	$T_{LE53} = (E_{132} - 515^* S - E_{78})/S$	"	"	"
	T_{LE54}	209	"	"	"	10 1111 1101	10 1111 11øø	"	E_{133}	"	$T_{LE54} = (E_{133} - 765^* S - E_{78})/S$	"	"	"
	T_{LE55}	210	"	"	"	10 1111 1110	10 1111 111ø	"	E_{134}	"	$T_{LE55} = (E_{134} - 766^* S - E_{78})/S$	"	"	"
	T_{LE56}	211	"	"	"	10 1111 1111	1ø øøøø øøøø	"	E_{135}	"	$T_{LE56} = (E_{135} - 767^* S - E_{78})/S$	"	"	"
	T_{LE57}	212	"	"	"	11 0000 0000	11 0000 000ø	"	E_{136}	"	$T_{LE57} = (E_{136} - 768^* S - E_{78})/S$	"	"	"
	T_{LE58}	213	"	"	"	11 0000 0001	11 0000 00øø	"	E_{137}	"	$T_{LE58} = (E_{137} - 769^* S - E_{78})/S$	"	"	"
	T_{LE59}	214	"	"	"	11 0000 0010	11 0000 001ø	"	E_{138}	"	$T_{LE59} = (E_{138} - 770^* S - E_{78})/S$	"	"	"
	T_{LE60}	215	"	"	"	11 0000 0011	11 0000 0øøø	"	E_{139}	"	$T_{LE60} = (E_{139} - 771^* S - E_{78})/S$	"	"	"
	T_{LE61}	216	"	"	"	11 0111 1101	11 0111 11øø	"	E_{140}	"	$T_{LE61} = (E_{140} - 893^* S - E_{78})/S$	"	"	"
	T_{LE62}	217	"	"	"	11 0111 1110	11 0111 111ø	"	E_{141}	"	$T_{LE62} = (E_{141} - 894^* S - E_{78})/S$	"	"	"
	T_{LE63}	218	"	"	"	11 0111 1111	11 øøøø øøøø	"	E_{142}	"	$T_{LE63} = (E_{142} - 895^* S - E_{78})/S$	"	"	"
	T_{LE64}	219	"	"	"	11 1000 0000	11 1000 000ø	"	E_{143}	"	$T_{LE64} = (E_{143} - 896^* S - E_{78})/S$	"	"	"
	T_{LE65}	220	"	"	"	11 1000 0001	11 1000 00øø	"	E_{144}	"	$T_{LE65} = (E_{144} - 897^* S - E_{78})/S$	"	"	"
	T_{LE66}	221	"	"	"	11 1000 0010	11 1000 001ø	"	E_{145}	"	$T_{LE66} = (E_{145} - 898^* S - E_{78})/S$	"	"	"
	T_{LE67}	222	"	"	"	11 1000 0011	11 1000 0øøø	"	E_{146}	"	$T_{LE67} = (E_{146} - 899^* S - E_{78})/S$	"	"	"
	T_{LE68}	223	"	"	"	11 1011 1101	11 1011 11øø	"	E_{147}	"	$T_{LE68} = (E_{147} - 957^* S - E_{78})/S$	"	"	"
	T_{LE69}	224	"	"	"	11 1011 1110	11 1011 111ø	"	E_{148}	"	$T_{LE69} = (E_{148} - 958^* S - E_{78})/S$	"	"	"
	T_{LE70}	225	"	"	"	11 1011 1111	11 1øøø øøøø	"	E_{149}	"	$T_{LE70} = (E_{149} - 959^* S - E_{78})/S$	"	"	"
	T_{LE71}	226	"	"	"	11 1100 0000	11 1100 000ø	"	E_{150}	"	$T_{LE71} = (E_{150} - 960^* S - E_{78})/S$	"	"	"
	T_{LE72}	227	"	"	"	11 1100 0001	11 1100 00øø	"	E_{151}	"	$T_{LE72} = (E_{151} - 961^* S - E_{78})/S$	"	"	"
	T_{LE73}	228	"	"	"	11 1100 0010	11 1100 001ø	"	E_{152}	"	$T_{LE73} = (E_{152} - 962^* S - E_{78})/S$	"	"	"
	T_{LE74}	229	"	"	"	11 1100 0011	11 1100 0øøø	"	E_{153}	"	$T_{LE74} = (E_{153} - 963^* S - E_{78})/S$	"	"	"
	T_{LE75}	230	"	"	"	11 1101 1101	11 1101 11øø	"	E_{154}	"	$T_{LE75} = (E_{154} - 989^* S - E_{78})/S$	"	"	"
	T_{LE76}	231	"	"	"	11 1101 1110	11 1101 111ø	"	E_{155}	"	$T_{LE76} = (E_{155} - 990^* S - E_{78})/S$	"	"	"
	T_{LE77}	232	"	"	"	11 1101 1111	11 11øø øøøø	"	E_{156}	"	$T_{LE77} = (E_{156} - 991^* S - E_{78})/S$	"	"	"
	T_{LE78}	233	"	"	"	11 1110 0000	11 1110 000ø	"	E_{157}	"	$T_{LE78} = (E_{157} - 992^* S - E_{78})/S$	"	"	"
	T_{LE79}	234	"	"	"	11 1110 0001	11 1110 00øø	"	E_{158}	"	$T_{LE79} = (E_{158} - 993^* S - E_{78})/S$	"	"	"
	T_{LE80}	235	"	"	"	11 1110 0010	11 1110 001ø	"	E_{159}	"	$T_{LE80} = (E_{159} - 994^* S - E_{78})/S$	"	"	"
	T_{LE81}	236	"	"	"	11 1110 0011	11 1110 0øøø	"	E_{160}	"	$T_{LE81} = (E_{160} - 995^* S - E_{78})/S$	"	"	"
	T_{LE82}	237	"	"	"	11 1110 1101	11 1110 11øø	"	E_{161}	"	$T_{LE82} = (E_{161} - 1005^* S - E_{78})/S$	"	"	"
	T_{LE83}	238	"	"	"	11 1110 1110	11 1110 111ø	"	E_{162}	"	$T_{LE83} = (E_{162} - 1006^* S - E_{78})/S$	"	"	"
	T_{LE84}	239	"	"	"	11 1110 1111	11 111ø øøøø	"	E_{163}	"	$T_{LE84} = (E_{163} - 1007^* S - E_{78})/S$	"	"	"
	T_{LE85}	240	"	"	"	11 1111 0000	11 1111 000ø	"	E_{164}	"	$T_{LE85} = (E_{164} - 1008^* S - E_{78})/S$	"	"	"
	T_{LE86}	241	"	"	"	11 1111 0001	11 1111 00øø	"	E_{165}	"	$T_{LE86} = (E_{165} - 1009^* S - E_{78})/S$	"	"	"
	T_{LE87}	242	"	"	"	11 1111 0010	11 1111 001ø	"	E_{166}	"	$T_{LE87} = (E_{166} - 1010^* S - E_{78})/S$	"	"	"

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltages ref. pin no. 14 (Figures 8, 9)			Applied digital code-word	Output digital code-word	Measurement sense lines ref. pin no. 14			Equations	Limits		Units
			10 V_{CC} (V)	12 V_{EE} (V)	15 BpC (V)	19<------>28 MSB<------>LSB	19<---->1,18,17 MSB<------>LSB	Pin	Value	Unit		Min	Max	
4 T_A = 25°C	T_{LE89}	243	+5.00	-15.00	0	11 1111 0011	11 1111 0ddd	13	E_{167}	V	$T_{LE88} = (E_{167} - 1011^* S - E_{78})/S$	-.50	+.50	LSB
	T_{LE89}	244	"	"	"	11 1111 0101	11 1111 01dd	"	E_{168}	"	$T_{LE89} = (E_{168} - 1013^* S - E_{78})/S$	"	"	"
	T_{LE90}	245	"	"	"	11 1111 0110	11 1111 011d	"	E_{169}	"	$T_{LE90} = (E_{169} - 1014^* S - E_{78})/S$	"	"	"
	T_{LE91}	246	"	"	"	11 1111 0111	11 1111 dddd	"	E_{170}	"	$T_{LE91} = (E_{170} - 1015^* S - E_{78})/S$	"	"	"
	T_{LE92}	247	"	"	"	11 1111 1000	11 1111 100d	"	E_{171}	"	$T_{LE92} = (E_{171} - 1016^* S - E_{78})/S$	"	"	"
	T_{LE93}	248	"	"	"	11 1111 1001	11 1111 10dd	"	E_{172}	"	$T_{LE93} = (E_{172} - 1017^* S - E_{78})/S$	"	"	"
	T_{LE94}	249	"	"	"	11 1111 1010	11 1111 101d	"	E_{173}	"	$T_{LE94} = (E_{173} - 1019^* S - E_{78})/S$	"	"	"
	T_{LE95}	250	"	"	"	11 1111 1011	11 1111 1ddd	"	E_{174}	"	$T_{LE95} = (E_{174} - 1019^* S - E_{78})/S$	"	"	"
	T_{LE96}	251	"	"	"	11 1111 1100	11 1111 110d	"	E_{175}	"	$T_{LE96} = (E_{175} - 1020^* S - E_{78})/S$	"	"	"
	T_{LE97}	252	"	"	"	11 1111 1101	11 1111 11dd	"	E_{176}	"	$T_{LE97} = (E_{176} - 1021^* S - E_{78})/S$	"	"	"
	T_{LE98}	253	"	"	"	11 1111 1110	11 1111 111d	"	E_{177}	"	$T_{LE98} = (E_{177} - 1022^* S - E_{78})/S$	"	"	"
											For differential linearity error, $D_{LE(N)} = (E(N+179) - E(N+178))/S$ where $S = (E_{178} - E_{179})/1022$ for N = 1 to 110			
			"	"	"	11 1111 1110	11 1111 111d	"	E_{178}	"				
			"	"	"	00 0000 0000	00 0000 000d	"	E_{179}	"				
	D_{LE1}	254	"	"	"	00 0000 0001	00 0000 00dd	"	E_{180}	"	$D_{LE1} = (E_{180} - E_{179})/S$	-0.90	+1.00	"
	D_{LE2}	255	"	"	"	00 0000 0010	00 0000 001d	"	E_{181}	"	$D_{LE2} = (E_{181} - E_{180})/S$	"	"	"
	D_{LE3}	256	"	"	"	00 0000 0011	00 0000 0ddd	"	E_{182}	"	$D_{LE3} = (E_{182} - E_{181})/S$	"	"	"
	D_{LE4}	257	"	"	"	00 0000 0100	00 0000 010d	"	E_{183}	"	$D_{LE4} = (E_{183} - E_{182})/S$	"	"	"
	D_{LE5}	258	"	"	"	00 0000 0101	00 0000 01dd	"	E_{184}	"	$D_{LE5} = (E_{184} - E_{183})/S$	"	"	"
	D_{LE6}	259	"	"	"	00 0000 0110	00 0000 011d	"	E_{185}	"	$D_{LE6} = (E_{185} - E_{184})/S$	"	"	"
	D_{LE7}	260	"	"	"	00 0000 0111	00 0000 dddd	"	E_{186}	"	$D_{LE7} = (E_{186} - E_{185})/S$	"	"	"
	D_{LE8}	261	"	"	"	00 0000 1000	00 0000 100d	"	E_{187}	"	$D_{LE8} = (E_{187} - E_{186})/S$	"	"	"
	D_{LE9}	262	"	"	"	00 0000 1001	00 0000 10dd	"	E_{188}	"	$D_{LE9} = (E_{188} - E_{187})/S$	"	"	"
	D_{LE10}	263	"	"	"	00 0000 1010	00 0000 101d	"	E_{189}	"	$D_{LE10} = (E_{189} - E_{188})/S$	"	"	"
	D_{LE11}	264	"	"	"	00 0000 1011	00 0000 1ddd	"	E_{190}	"	$D_{LE11} = (E_{190} - E_{189})/S$	"	"	"
	D_{LE12}	265	"	"	"	00 0000 1100	00 0000 110d	"	E_{191}	"				
	D_{LE13}	266	"	"	"	00 0000 1101	00 0000 11dd	"	E_{192}	"	$D_{LE13} = (E_{192} - E_{191})/S$	"	"	"
	D_{LE14}	267	"	"	"	00 0000 1110	00 0000 111d	"	E_{193}	"	$D_{LE14} = (E_{193} - E_{192})/S$	"	"	"
	D_{LE15}	268	"	"	"	00 0000 1111	00 000d dddd	"	E_{194}	"	$D_{LE15} = (E_{194} - E_{193})/S$	"	"	"
	D_{LE16}	269	"	"	"	00 0001 0000	00 0001 000d	"	E_{195}	"	$D_{LE16} = (E_{195} - E_{194})/S$	"	"	"
	D_{LE17}	270	"	"	"	00 0001 0001	00 0001 00dd	"	E_{196}	"	$D_{LE17} = (E_{196} - E_{195})/S$	"	"	"
	D_{LE18}	271	"	"	"	00 0001 0010	00 0001 001d	"	E_{197}	"	$D_{LE18} = (E_{197} - E_{196})/S$	"	"	"
	D_{LE19}	272	"	"	"	00 0001 0011	00 0001 0ddd	"	E_{198}	"	$D_{LE19} = (E_{198} - E_{197})/S$	"	"	"
	D_{LE20}	273	"	"	"	00 0001 1100	00 0001 110d	"	E_{199}	"				
	D_{LE21}	274	"	"	"	00 0001 1101	00 0001 11dd	"	E_{200}	"	$D_{LE21} = (E_{200} - E_{199})/S$	"	"	"
	D_{LE22}	275	"	"	"	00 0001 1110	00 0001 111d	"	E_{201}	"	$D_{LE22} = (E_{201} - E_{200})/S$	"	"	"
	D_{LE23}	276	"	"	"	00 0001 1111	00 000d dddd	"	E_{202}	"	$D_{LE23} = (E_{202} - E_{201})/S$	"	"	"
	D_{LE24}	277	"	"	"	00 0010 0000	00 0010 000d	"	E_{203}	"	$D_{LE24} = (E_{203} - E_{202})/S$	"	"	"
	D_{LE25}	278	"	"	"	00 0010 0001	00 0010 00dd	"	E_{204}	"	$D_{LE25} = (E_{204} - E_{203})/S$	"	"	"
	D_{LE26}	279	"	"	"	00 0010 0010	00 0010 001d	"	E_{205}	"	$D_{LE26} = (E_{205} - E_{204})/S$	"	"	"
	D_{LE27}	280	"	"	"	00 0010 0011	00 0010 0ddd	"	E_{206}	"	$D_{LE27} = (E_{206} - E_{205})/S$	"	"	"
	D_{LE28}	281	"	"	"	00 0011 1100	00 0011 110d	"	E_{207}	"				
	D_{LE29}	282	"	"	"	00 0011 1101	00 0011 11dd	"	E_{208}	"	$D_{LE29} = (E_{208} - E_{207})/S$	"	"	"
	D_{LE30}	283	"	"	"	00 0011 1110	00 0011 111d	"	E_{209}	"	$D_{LE30} = (E_{209} - E_{208})/S$	"	"	"
	D_{LE31}	284	"	"	"	00 0011 1111	00 0ddd dddd	"	E_{210}	"	$D_{LE31} = (E_{210} - E_{209})/S$	"	"	"
	D_{LE32}	285	"	"	"	00 0100 0000	00 0100 000d	"	E_{211}	"	$D_{LE32} = (E_{211} - E_{210})/S$	"	"	"
	D_{LE33}	286	"	"	"	00 0100 0001	00 0100 00dd	"	E_{212}	"	$D_{LE33} = (E_{212} - E_{211})/S$	"	"	"
	D_{LE34}	287	"	"	"	00 0100 0010	00 0100 001d	"	E_{213}	"	$D_{LE34} = (E_{213} - E_{212})/S$	"	"	"

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltages ref. pin no. 14 (Figures 8, 9) 10 V_{CC} (V)	12 V_{EE} (V)	15 BpC (V)	Applied digital code-word 19<------>28 MSB<------>LSB	Output digital code-word 19<----->1,18,17 MSB<------->LSB	Measurement sense lines ref. pin no. 14 Pin	Value	Unit	Equations	Limits Min	Max	Units
4 T_A = 25°C	D_{LE35}	288	+5.00	-15.00	0	00 0100 0011	00 0100 0ØØØ	13	E_{214}	V	$D_{LE35} = (E_{214} - E_{213})/S$	-0.90	+1.00	LSB
	D_{LE36}	289	"	"	"	00 0111 1100	00 0111 110Ø	"	E_{215}	"				
	D_{LE37}	290	"	"	"	00 0111 1101	00 0111 11ØØ	"	E_{216}	"	$D_{LE37} = (E_{216} - E_{215})/S$	"	"	"
	D_{LE38}	291	"	"	"	00 0111 1110	00 0111 111Ø	"	E_{217}	"	$D_{LE38} = (E_{217} - E_{216})/S$	"	"	"
	D_{LE39}	292	"	"	"	00 0111 1111	00 ØØØØ ØØØØ	"	E_{218}	"	$D_{LE39} = (E_{218} - E_{217})/S$	"	"	"
	D_{LE40}	293	"	"	"	00 1000 0000	00 1000 000Ø	"	E_{219}	"	$D_{LE40} = (E_{219} - E_{218})/S$	"	"	"
	D_{LE41}	294	"	"	"	00 1000 0001	00 1000 00ØØ	"	E_{220}	"	$D_{LE41} = (E_{220} - E_{219})/S$	"	"	"
	D_{LE42}	295	"	"	"	00 1000 0010	00 1000 001Ø	"	E_{221}	"	$D_{LE42} = (E_{221} - E_{220})/S$	"	"	"
	D_{LE43}	296	"	"	"	00 1000 0011	00 1000 0ØØØ	"	E_{222}	"	$D_{LE43} = (E_{222} - E_{221})/S$	"	"	"
	D_{LE44}	297	"	"	"	00 1111 1100	00 1111 110Ø	"	E_{223}	"				
	D_{LE45}	298	"	"	"	00 1111 1101	00 1111 11ØØ	"	E_{224}	"	$D_{LE45} = (E_{224} - E_{223})/S$	"	"	"
	D_{LE46}	299	"	"	"	00 1111 1110	00 1111 111Ø	"	E_{225}	"	$D_{LE46} = (E_{225} - E_{224})/S$	"	"	"
	D_{LE47}	300	"	"	"	00 1111 1111	0Ø ØØØØ ØØØØ	"	E_{226}	"	$D_{LE47} = (E_{226} - E_{225})/S$	"	"	"
	D_{LE48}	301	"	"	"	01 0000 0000	01 0000 000Ø	"	E_{227}	"	$D_{LE48} = (E_{227} - E_{226})/S$	"	"	"
	D_{LE49}	302	"	"	"	01 0000 0001	01 0000 00ØØ	"	E_{228}	"	$D_{LE49} = (E_{228} - E_{227})/S$	"	"	"
	D_{LE50}	303	"	"	"	01 0000 0010	01 0000 001Ø	"	E_{229}	"	$D_{LE50} = (E_{229} - E_{228})/S$	"	"	"
	D_{LE51}	304	"	"	"	01 0000 0011	01 0000 0ØØØ	"	E_{230}	"	$D_{LE51} = (E_{230} - E_{229})/S$	"	"	"
	D_{LE52}	305	"	"	"	01 1111 1100	01 1111 110Ø	"	E_{231}	"				
	D_{LE53}	306	"	"	"	01 1111 1101	01 1111 11ØØ	"	E_{232}	"	$D_{LE53} = (E_{232} - E_{231})/S$	"	"	"
	D_{LE54}	307	"	"	"	01 1111 1110	01 1111 111Ø	"	E_{233}	"	$D_{LE54} = (E_{233} - E_{232})/S$	"	"	"
	D_{LE55}	308	"	"	"	01 1111 1111	ØØ ØØØØ ØØØØ	"	E_{234}	"	$D_{LE55} = (E_{234} - E_{233})/S$	"	"	"
	D_{LE56}	309	"	"	"	10 0000 0000	10 0000 000Ø	"	E_{235}	"	$D_{LE56} = (E_{235} - E_{234})/S$	"	"	"
	D_{LE57}	310	"	"	"	10 0000 0001	10 0000 00ØØ	"	E_{236}	"	$D_{LE57} = (E_{236} - E_{235})/S$	"	"	"
	D_{LE58}	311	"	"	"	10 0000 0010	10 0000 001Ø	"	E_{237}	"	$D_{LE58} = (E_{237} - E_{236})/S$	"	"	"
	D_{LE59}	312	"	"	"	10 0000 0011	10 0000 0ØØØ	"	E_{238}	"	$D_{LE59} = (E_{238} - E_{237})/S$	"	"	"
	D_{LE60}	313	"	"	"	10 1111 1100	10 1111 110Ø	"	E_{239}	"				
	D_{LE61}	314	"	"	"	10 1111 1101	10 1111 11ØØ	"	E_{240}	"	$D_{LE61} = (E_{240} - E_{239})/S$	"	"	"
	D_{LE62}	315	"	"	"	10 1111 1110	10 1111 111Ø	"	E_{241}	"	$D_{LE62} = (E_{241} - E_{240})/S$	"	"	"
	D_{LE63}	316	"	"	"	10 1111 1111	1Ø ØØØØ ØØØØ	"	E_{242}	"	$D_{LE63} = (E_{242} - E_{241})/S$	"	"	"
	D_{LE64}	317	"	"	"	11 0000 0000	11 0000 000Ø	"	E_{243}	"	$D_{LE64} = (E_{243} - E_{242})/S$	"	"	"
	D_{LE65}	318	"	"	"	11 0000 0001	11 0000 00ØØ	"	E_{244}	"	$D_{LE65} = (E_{244} - E_{243})/S$	"	"	"
	D_{LE66}	319	"	"	"	11 0000 0010	11 0000 001Ø	"	E_{245}	"	$D_{LE66} = (E_{245} - E_{244})/S$	"	"	"
	D_{LE67}	320	"	"	"	11 0000 0011	11 0000 0ØØØ	"	E_{246}	"	$D_{LE67} = (E_{246} - E_{245})/S$	"	"	"
	D_{LE68}	321	"	"	"	11 0111 1100	11 0111 110Ø	"	E_{247}	"				
	D_{LE69}	322	"	"	"	11 0111 1101	11 0111 11ØØ	"	E_{248}	"	$D_{LE69} = (E_{248} - E_{247})/S$	"	"	"
	D_{LE70}	323	"	"	"	11 0111 1110	11 0111 111Ø	"	E_{249}	"	$D_{LE70} = (E_{249} - E_{248})/S$	"	"	"
	D_{LE71}	324	"	"	"	11 0111 1111	11 ØØØØ ØØØØ	"	E_{250}	"	$D_{LE71} = (E_{250} - E_{249})/S$	"	"	"
	D_{LE72}	325	"	"	"	11 1000 0000	11 1000 000Ø	"	E_{251}	"	$D_{LE72} = (E_{251} - E_{250})/S$	"	"	"
	D_{LE73}	326	"	"	"	11 1000 0001	11 1000 00ØØ	"	E_{252}	"	$D_{LE73} = (E_{252} - E_{251})/S$	"	"	"
	D_{LE74}	327	"	"	"	11 1000 0010	11 1000 001Ø	"	E_{253}	"	$D_{LE74} = (E_{253} - E_{252})/S$	"	"	"
	D_{LE75}	328	"	"	"	11 1000 0011	11 1000 0ØØØ	"	E_{254}	"	$D_{LE75} = (E_{254} - E_{253})/S$	"	"	"
	D_{LE76}	329	"	"	"	11 1011 1100	11 1011 110Ø	"	E_{255}	"				
	D_{LE77}	330	"	"	"	11 1011 1101	11 1011 11ØØ	"	E_{256}	"	$D_{LE77} = (E_{256} - E_{255})/S$	"	"	"
	D_{LE78}	331	"	"	"	11 1011 1110	11 1011 111Ø	"	E_{257}	"	$D_{LE78} = (E_{257} - E_{256})/S$	"	"	"
	D_{LE79}	332	"	"	"	11 1011 1111	11 1ØØØ ØØØØ	"	E_{258}	"	$D_{LE79} = (E_{258} - E_{257})/S$	"	"	"
	D_{LE80}	333	"	"	"	11 1100 0000	11 1100 000Ø	"	E_{259}	"	$D_{LE80} = (E_{259} - E_{258})/S$	"	"	"
	D_{LE81}	334	"	"	"	11 1100 0001	11 1100 00ØØ	"	E_{260}	"	$D_{LE81} = (E_{260} - E_{259})/S$	"	"	"
	D_{LE82}	335	"	"	"	11 1100 0010	11 1100 001Ø	"	E_{261}	"	$D_{LE82} = (E_{261} - E_{260})/S$	"	"	"
	D_{LE83}	336	"	"	"	11 1100 0011	11 1100 0ØØØ	"	E_{262}	"	$D_{LE83} = (E_{262} - E_{261})/S$	"	"	"
	D_{LE84}	337	"	"	"	11 1101 1100	11 1101 110Ø	"	E_{263}	"				
	D_{LE85}	338	"	"	"	11 1101 1101	11 1101 11ØØ	"	E_{264}	"	$D_{LE85} = (E_{264} - E_{263})/S$	"	"	"
	D_{LE86}	339	"	"	"	11 1101 1110	11 1101 111Ø	"	E_{265}	"	$D_{LE86} = (E_{265} - E_{264})/S$	"	"	"

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltages ref. pin no. 14 (Figures 8, 9) 10 V_{CC} (V)	12 V_{EE} (V)	15 BpC (V)	Applied digital code-word 19<------>2A MSB<------>LSB	Output digital code-word 9<----->1,18,17 MSB<------>LSB	Measurement sense lines ref. pin no. 14 Pin	Value	Unit	Equations	Limits Min	Limits Max	Units
4 T_A = 25°C	D_{LE87}	340	+5.00	-15.00	0	11 1101 1111	11 11dd dddd	13	E_{266}	V	$D_{LE87} = (E_{266} - E_{265})/S$	-0.90	+1.00	LSB
	D_{LE88}	341	"	"	"	11 1110 0000	11 1110 000d	"	E_{267}	"	$D_{LE88} = (E_{267} - E_{266})/S$	"	"	"
	D_{LE89}	342	"	"	"	11 1110 0001	11 1110 00dd	"	E_{268}	"	$D_{LE89} = (E_{268} - E_{267})/S$	"	"	"
	D_{LE90}	343	"	"	"	11 1110 0010	11 1110 001d	"	E_{269}	"	$D_{LE90} = (E_{269} - E_{268})/S$	"	"	"
	D_{LE91}	344	"	"	"	11 1110 0011	11 1110 0ddd	"	E_{270}	"	$D_{LE91} = (E_{270} - E_{269})/S$	"	"	"
	D_{LE92}	345	"	"	"	11 1110 1100	11 1110 110d	"	E_{271}	"				
	D_{LE93}	346	"	"	"	11 1110 1101	11 1110 11dd	"	E_{272}	"	$D_{LE93} = (E_{272} - E_{271})/S$	"	"	"
	D_{LE94}	347	"	"	"	11 1110 1110	11 1110 111d	"	E_{273}	"	$D_{LE94} = (E_{273} - E_{272})/S$	"	"	"
	D_{LE95}	348	"	"	"	11 1110 1111	11 111d dddd	"	E_{274}	"	$D_{LE95} = (E_{274} - E_{273})/S$	"	"	"
	D_{LE96}	349	"	"	"	11 1111 0000	11 1111 000d	"	E_{275}	"	$D_{LE96} = (E_{275} - E_{274})/S$	"	"	"
	D_{LE97}	350	"	"	"	11 1111 0001	11 1111 00dd	"	E_{276}	"	$D_{LE97} = (E_{276} - E_{275})/S$	"	"	"
	D_{LE98}	351	"	"	"	11 1111 0010	11 1111 001d	"	E_{277}	"	$D_{LE98} = (E_{277} - E_{276})/S$	"	"	"
	D_{LE99}	352	"	"	"	11 1111 0011	11 1111 0ddd	"	E_{278}	"	$D_{LE99} = (E_{278} - E_{277})/S$	"	"	"
	D_{LE100}	353	"	"	"	11 1111 0100	11 1111 010d	"	E_{279}	"				
	D_{LE101}	354	"	"	"	11 1111 0101	11 1111 01dd	"	E_{280}	"	$D_{LE101} = (E_{280} - E_{279})/S$	"	"	"
	D_{LE102}	355	"	"	"	11 1111 0110	11 1111 011d	"	E_{281}	"	$D_{LE102} = (E_{281} - E_{280})/S$	"	"	"
	D_{LE103}	356	"	"	"	11 1111 0111	11 1111 dddd	"	E_{282}	"	$D_{LE103} = (E_{282} - E_{281})/S$	"	"	"
	D_{LE104}	357	"	"	"	11 1111 1000	11 1111 100d	"	E_{283}	"	$D_{LE104} = (E_{283} - E_{282})/S$	"	"	"
	D_{LE105}	358	"	"	"	11 1111 1001	11 1111 10dd	"	E_{284}	"	$D_{LE105} = (E_{284} - E_{283})/S$	"	"	"
	D_{LE106}	359	"	"	"	11 1111 1010	11 1111 101d	"	E_{285}	"	$D_{LE106} = (E_{285} - E_{284})/S$	"	"	"
	D_{LE107}	360	"	"	"	11 1111 1011	11 1111 1ddd	"	E_{286}	"	$D_{LE107} = (E_{286} - E_{285})/S$	"	"	"
	D_{LE108}	361	"	"	"	11 1111 1100	11 1111 110d	"	E_{287}	"	$D_{LE108} = (E_{287} - E_{286})/S$	"	"	"
	D_{LE109}	362	"	"	"	11 1111 1101	11 1111 11dd	"	E_{288}	"	$D_{LE109} = (E_{288} - E_{287})/S$	"	"	"
	D_{LE110}	363	"	"	"	11 1111 1110	11 1111 111d	"	E_{289}	"	$D_{LE110} = (E_{289} - E_{288})/S$	"	"	"
			+5.00	-15.00	0	11 1111 1110	11 1111 111d	"	E_{290}	"	For S = $(E_{290} - E_{291})/1022$			
			"	"	"	00 0000 0000	00 0000 000d	"	E_{291}	"				
	$-P_{SSa}$	364	"	-13.50	"	" " "	" " "	"	E_{292}	"	$-P_{SSa} = (E_{292} - E_{291})/(10.00*S)$	-0.12	0.12	LSB/%PS
			"	-16.00	"	" " "	" " "	"	E_{293}	"	$-P_{SSa} = (E_{291} - E_{293})/(6.67*S)$	"	"	"
	$-P_{SSb}$	365	"	-13.50	"	11 1111 1110	11 1111 111d	"	E_{294}	"	$-P_{SSb} = (E_{294} - E_{290})/(10.00*S)$	"	"	"
			"	-16.00	"	" " "	" " "	"	E_{295}	"	$-P_{SSb} = (E_{290} - E_{295})/(6.67*S)$	"	"	"
			+5.00	-15.00	0	11 1111 1110	11 1111 111d	"	E_{296}	"	For S = $(E_{296} - E_{297})/1022$			
			"	"	"	00 0000 0000	00 0000 000d	"	E_{297}	"				
	$+P_{SSa}$	366	+5.50	"	"	" " "	" " "	"	E_{298}	"	$+P_{SSa} = (E_{298} - E_{297})/(10.00*S)$	-0.1	0.1	"
			+4.50	"	"	" " "	" " "	"	E_{299}	"	$+P_{SSa} = (E_{297} - E_{299})/(10.00*S)$	"	"	"
	$+P_{SSb}$	367	+5.50	"	"	11 1111 1110	11 1111 111d	"	E_{300}	"	$+P_{SSb} = (E_{300} - E_{296})/(10.00*S)$	"	"	"
			+4.50	"	"	" " "	" " "	"	E_{301}	"	$+P_{SSb} = (E_{296} - E_{301})/(10.00*S)$	"	"	"
5 T_A = 125°C	Z_E	368	+5.00	-15.00	0	00 0000 0000	00 0000 000d	"	E_{302}	"	$Z_E = 102.4* (E_{302})$	-3	+3	LSB
	R_{ZF}	369	"	"	Open	01 1111 1111	dd dddd dddd	"	E_{303}	"	$Z_E = 102.4* (E_{303})$	"	"	"
	$-V_{FSE}$	370	"	"	Open	00 0000 0000	00 0000 000d	"	E_{304}	"	$-V_{FSE} = 102.4* (E_{304} +5 +10/1024)$	-8.7	8.7	"
•	$+V_{FSE}$	371	"	"	0	11 1111 1110	11 1111 111d	"	E_{305}	"	$+V_{FSE} = 102.4* (E_{305} -10 -10/1024)$	"	"	"
		372	"	"	0	" " "	" " "	"	E_{306}	"	$+V_{FSE} = 102.4* (E_{306} -5 -10/1024)$	"	"	"

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TABLE III. Group A Inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltages ref. pin no. 14 (Figures 8, 9) 10 V_{CC} (V)	12 V_{EE} (V)	15 BpC (V)	Applied digital code-word 19<------>28 MSB<------>LSB	Output digital code-word 9<----->1,18,17 MSB<------>LSB	Measurement sense lines ref. pin no. 14 Pin	Value	Unit	Equations	Limits Min	Limits Max	Units
5 T_A = 125°C											For end-point linearity error, TLE(N)=(E(N+308)-V(ideal))/S where S = (E_{308} - E_{307})/1022 for N = 1 to 98			
			+5.00	-15.00	0	00 0000 0000	00 0000 000ϕ	13	E_{307}	V				
			"	"	"	11 1111 1110	11 1111 111ϕ	"	E_{308}	"				
	T_{LE1}	373	"	"	"	00 0000 0001	00 0000 00ϕϕ	"	E_{309}	"	T_{LE1} = (E_{309} - 1* S - E_{78})/S	-.50	+.50	LSB
	T_{LE2}	374	"	"	"	00 0000 0010	00 0000 001ϕ	"	E_{310}	"	T_{LE2} = (E_{310} - 2* S - E_{78})/S	"	"	"
	T_{LE3}	375	"	"	"	00 0000 0011	00 0000 0ϕϕϕ	"	E_{311}	"	T_{LE3} = (E_{311} - 3* S - E_{78})/S	"	"	"
	T_{LE4}	376	"	"	"	00 0000 0100	00 0000 010ϕ	"	E_{312}	"	T_{LE4} = (E_{312} - 4* S - E_{78})/S	"	"	"
	T_{LE5}	377	"	"	"	00 0000 0101	00 0000 01ϕϕ	"	E_{313}	"	T_{LE5} = (E_{313} - 5* S - E_{78})/S	"	"	"
	T_{LE6}	378	"	"	"	00 0000 0110	00 0000 011ϕ	"	E_{314}	"	T_{LE6} = (E_{314} - 6* S - E_{78})/S	"	"	"
	T_{LE7}	379	"	"	"	00 0000 0111	00 0000 ϕϕϕϕ	"	E_{315}	"	T_{LE7} = (E_{315} - 7* S - E_{78})/S	"	"	"
	T_{LE8}	380	"	"	"	00 0000 1000	00 0000 100ϕ	"	E_{316}	"	T_{LE8} = (E_{316} - 8* S - E_{78})/S	"	"	"
	T_{LE9}	381	"	"	"	00 0000 1001	00 0000 10ϕϕ	"	E_{317}	"	T_{LE9} = (E_{317} - 9* S - E_{78})/S	"	"	"
	T_{LE10}	382	"	"	"	00 0000 1010	00 0000 101ϕ	"	E_{318}	"	T_{LE10} = (E_{318} - 10* S - E_{78})/S	"	"	"
	T_{LE11}	383	"	"	"	00 0000 1011	00 0000 1ϕϕϕ	"	E_{319}	"	T_{LE11} = (E_{319} - 11* S - E_{78})/S	"	"	"
	T_{LE12}	384	"	"	"	00 0000 1101	00 0000 110ϕ	"	E_{320}	"	T_{LE12} = (E_{320} - 13* S - E_{78})/S	"	"	"
	T_{LE13}	385	"	"	"	00 0000 1110	00 0000 111ϕ	"	E_{321}	"	T_{LE13} = (E_{321} - 14* S - E_{78})/S	"	"	"
	T_{LE14}	386	"	"	"	00 0000 1111	00 000ϕ ϕϕϕϕ	"	E_{322}	"	T_{LE14} = (E_{322} - 15* S - E_{78})/S	"	"	"
	T_{LE15}	387	"	"	"	00 0001 0000	00 0001 000ϕ	"	E_{323}	"	T_{LE15} = (E_{323} - 16* S - E_{78})/S	"	"	"
	T_{LE16}	388	"	"	"	00 0001 0001	00 0001 00ϕϕ	"	E_{324}	"	T_{LE16} = (E_{324} - 17* S - E_{78})/S	"	"	"
	T_{LE17}	389	"	"	"	00 0001 0010	00 0001 001ϕ	"	E_{325}	"	T_{LE17} = (E_{325} - 18* S - E_{78})/S	"	"	"
	T_{LE18}	390	"	"	"	00 0001 0011	00 0001 0ϕϕϕ	"	E_{326}	"	T_{LE18} = (E_{326} - 19* S - E_{78})/S	"	"	"
	T_{LE19}	391	"	"	"	00 0001 1101	00 0001 110ϕ	"	E_{327}	"	T_{LE19} = (E_{327} - 29* S - E_{78})/S	"	"	"
	T_{LE20}	392	"	"	"	00 0001 1110	00 0001 111ϕ	"	E_{328}	"	T_{LE20} = (E_{328} - 30* S - E_{78})/S	"	"	"
	T_{LE21}	393	"	"	"	00 0001 1111	00 00ϕϕ ϕϕϕϕ	"	E_{329}	"	T_{LE21} = (E_{329} - 31* S - E_{78})/S	"	"	"
	T_{LE22}	394	"	"	"	00 0010 0000	00 0010 000ϕ	"	E_{330}	"	T_{LE22} = (E_{330} - 32* S - E_{78})/S	"	"	"
	T_{LE23}	395	"	"	"	00 0010 0001	00 0010 00ϕϕ	"	E_{331}	"	T_{LE23} = (E_{331} - 33* S - E_{78})/S	"	"	"
	T_{LE24}	396	"	"	"	00 0010 0010	00 0010 001ϕ	"	E_{332}	"	T_{LE24} = (E_{332} - 34* S - E_{78})/S	"	"	"
	T_{LE25}	397	"	"	"	00 0010 0011	00 0010 0ϕϕϕ	"	E_{333}	"	T_{LE25} = (E_{333} - 35* S - E_{78})/S	"	"	"
	T_{LE26}	398	"	"	"	00 0011 1101	00 0011 110ϕ	"	E_{334}	"	T_{LE26} = (E_{334} - 61* S - E_{78})/S	"	"	"
	T_{LE27}	399	"	"	"	00 0011 1110	00 0011 111ϕ	"	E_{335}	"	T_{LE27} = (E_{335} - 62* S - E_{78})/S	"	"	"
	T_{LE28}	400	"	"	"	00 0011 1111	00 0ϕϕϕ ϕϕϕϕ	"	E_{336}	"	T_{LE28} = (E_{336} - 63* S - E_{78})/S	"	"	"
	T_{LE29}	401	"	"	"	00 0100 0000	00 0100 000ϕ	"	E_{337}	"	T_{LE29} = (E_{337} - 64* S - E_{78})/S	"	"	"
	T_{LE30}	402	"	"	"	00 0100 0001	00 0100 00ϕϕ	"	E_{338}	"	T_{LE30} = (E_{338} - 65* S - E_{78})/S	"	"	"
	T_{LE31}	403	"	"	"	00 0100 0010	00 0100 001ϕ	"	E_{339}	"	T_{LE31} = (E_{339} - 66* S - E_{78})/S	"	"	"
	T_{LE32}	404	"	"	"	00 0100 0011	00 0100 0ϕϕϕ	"	E_{340}	"	T_{LE32} = (E_{340} - 67* S - E_{78})/S	"	"	"
	T_{LE33}	405	"	"	"	00 0111 1101	00 0111 110ϕ	"	E_{341}	"	T_{LE33} = (E_{341} - 125* S - E_{78})/S	"	"	"
	T_{LE34}	406	"	"	"	00 0111 1110	00 0111 111ϕ	"	E_{342}	"	T_{LE34} = (E_{342} - 126* S - E_{78})/S	"	"	"
	T_{LE35}	407	"	"	"	00 0111 1111	00 ϕϕϕϕ ϕϕϕϕ	"	E_{343}	"	T_{LE35} = (E_{343} - 127* S - E_{78})/S	"	"	"
	T_{LE36}	408	"	"	"	00 1000 0000	00 1000 000ϕ	"	E_{344}	"	T_{LE36} = (E_{344} - 128* S - E_{78})/S	"	"	"
	T_{LE37}	409	"	"	"	00 1000 0001	00 1000 00ϕϕ	"	E_{345}	"	T_{LE37} = (E_{345} - 129* S - E_{78})/S	"	"	"
	T_{LE38}	410	"	"	"	00 1000 0010	00 1000 001ϕ	"	E_{346}	"	T_{LE38} = (E_{346} - 130* S - E_{78})/S	"	"	"
	T_{LE39}	411	"	"	"	00 1000 0011	00 1000 0ϕϕϕ	"	E_{347}	"	T_{LE39} = (E_{347} - 131* S - E_{78})/S	"	"	"
	T_{LE40}	412	"	"	"	00 1111 1101	00 1111 110ϕ	"	E_{348}	"	T_{LE40} = (E_{348} - 253* S - E_{78})/S	"	"	"
	T_{LE41}	413	"	"	"	00 1111 1110	00 1111 111ϕ	"	E_{349}	"	T_{LE41} = (E_{349} - 254* S - E_{78})/S	"	"	"
	T_{LE42}	414	"	"	"	00 1111 1111	00 ϕϕϕϕ ϕϕϕϕ	"	E_{350}	"	T_{LE42} = (E_{350} - 255* S - E_{78})/S	"	"	"
	T_{LE43}	415	"	"	"	01 0000 0000	01 0000 000ϕ	"	E_{351}	"	T_{LE43} = (E_{351} - 256* S - E_{78})/S	"	"	"
	T_{LE44}	416	"	"	"	01 0000 0001	01 0000 00ϕϕ	"	E_{352}	"	T_{LE44} = (E_{352} - 257* S - E_{78})/S	"	"	"
	T_{LE45}	417	"	"	"	01 0000 0010	01 0000 001ϕ	"	E_{353}	"	T_{LE45} = (E_{353} - 258* S - E_{78})/S	"	"	"

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltages ref. pin no. 14 (figures 8, 9)			Applied digital code-word	Output digital code-word	Measurement sense lines ref. pin no. 14			Equations	Limits		Units
			13 V_{CC} (V)	12 V_{EE} (V)	15 BpC (V)	19<------>28 MSB<------>LSB	9<---->1,18,17 MSB<------>LSB	Pin	Value	Unit		Min	Max	
5, T_A = 125°C	TLE46	418	+5.00	-15.00	0	01 0000 0011	01 0000 0666	13	E354	V	TLE46 = (E354 - 259* S - E78)/S	-.50	+.50	LSB
	TLE47	419	"	"	"	01 1111 1101	01 1111 1166	"	E355	"	TLE47 = (E355 - 509* S - E78)/S	"	"	"
	TLE48	420	"	"	"	01 1111 1110	01 1111 1116	"	E356	"	TLE48 = (E356 - 510* S - E78)/S	"	"	"
	TLE49	421	"	"	"	01 1111 1111	01 6666 6666	"	E357	"	TLE49 = (E357 - 511* S - E78)/S	"	"	"
	TLE50	422	"	"	"	10 0000 0000	10 0000 0006	"	E358	"	TLE50 = (E358 - 512* S - E78)/S	"	"	"
	TLE51	423	"	"	"	10 0000 0001	10 0000 0066	"	E359	"	TLE51 = (E359 - 513* S - E78)/S	"	"	"
	TLE52	424	"	"	"	10 0000 0010	10 0000 0016	"	E360	"	TLE52 = (E360 - 514* S - E78)/S	"	"	"
	TLE53	425	"	"	"	10 0000 0011	10 0000 0666	"	E361	"	TLE53 = (E361 - 515* S - E78)/S	"	"	"
	TLE54	426	"	"	"	10 1111 1101	10 1111 1166	"	E362	"	TLE54 = (E362 - 765* S - E78)/S	"	"	"
	TLE55	427	"	"	"	10 1111 1110	10 1111 1116	"	E363	"	TLE55 = (E363 - 766* S - E78)/S	"	"	"
	TLE56	428	"	"	"	10 1111 1111	10 6666 6666	"	E364	"	TLE56 = (E364 - 767* S - E78)/S	"	"	"
	TLE57	429	"	"	"	11 0000 0000	11 0000 0006	"	E365	"	TLE57 = (E365 - 768* S - E78)/S	"	"	"
	TLE58	430	"	"	"	11 0000 0001	11 0000 0066	"	E366	"	TLE58 = (E366 - 769* S - E78)/S	"	"	"
	TLE59	431	"	"	"	11 0000 0010	11 0000 0016	"	E367	"	TLE59 = (E367 - 770* S - E78)/S	"	"	"
	TLE60	432	"	"	"	11 0000 0011	11 0000 0666	"	E368	"	TLE60 = (E368 - 771* S - E78)/S	"	"	"
	TLE61	433	"	"	"	11 0111 1101	11 0111 1166	"	E369	"	TLE61 = (E369 - 893* S - E78)/S	"	"	"
	TLE62	434	"	"	"	11 0111 1110	11 0111 1116	"	E370	"	TLE62 = (E370 - 894* S - E78)/S	"	"	"
	TLE63	435	"	"	"	11 0111 1111	11 0111 1166	"	E371	"	TLE63 = (E371 - 895* S - E78)/S	"	"	"
	TLE64	436	"	"	"	11 1000 0000	11 1000 0006	"	E372	"	TLE64 = (E372 - 896* S - E78)/S	"	"	"
	TLE65	437	"	"	"	11 1000 0001	11 1000 0066	"	E373	"	TLE65 = (E373 - 897* S - E78)/S	"	"	"
	TLE66	438	"	"	"	11 1000 0010	11 1000 0016	"	E374	"	TLE66 = (E374 - 898* S - E78)/S	"	"	"
	TLE67	439	"	"	"	11 1000 0011	11 1000 0666	"	E375	"	TLE67 = (E375 - 899* S - E78)/S	"	"	"
	TLE68	440	"	"	"	11 1011 1101	11 1011 1166	"	E376	"	TLE68 = (E376 - 957* S - E78)/S	"	"	"
	TLE69	441	"	"	"	11 1011 1110	11 1011 1116	"	E377	"	TLE69 = (E377 - 958* S - E78)/S	"	"	"
	TLE70	442	"	"	"	11 1011 1111	11 1666 6666	"	E378	"	TLE70 = (E378 - 959* S - E78)/S	"	"	"
	TLE71	443	"	"	"	11 1100 0000	11 1100 0006	"	E379	"	TLE71 = (E379 - 960* S - E78)/S	"	"	"
	TLE72	444	"	"	"	11 1100 0001	11 1100 0066	"	E380	"	TLE72 = (E380 - 961* S - E78)/S	"	"	"
	TLE73	445	"	"	"	11 1100 0010	11 1100 0016	"	E381	"	TLE73 = (E381 - 962* S - E78)/S	"	"	"
	TLE74	446	"	"	"	11 1100 0011	11 1100 0666	"	E382	"	TLE74 = (E382 - 963* S - E78)/S	"	"	"
	TLE75	447	"	"	"	11 1101 1101	11 1101 1166	"	E383	"	TLE75 = (E383 - 980* S - E78)/S	"	"	"
	TLE76	448	"	"	"	11 1101 1110	11 1101 1116	"	E384	"	TLE76 = (E384 - 990* S - E78)/S	"	"	"
	TLE77	449	"	"	"	11 1101 1111	11 1666 6666	"	E385	"	TLE77 = (E385 - 991* S - E78)/S	"	"	"
	TLE78	450	"	"	"	11 1110 0000	11 1110 0006	"	E386	"	TLE78 = (E386 - 992* S - E78)/S	"	"	"
	TLE79	451	"	"	"	11 1110 0001	11 1110 0066	"	E387	"	TLE79 = (E387 - 993* S - E78)/S	"	"	"
	TLE80	452	"	"	"	11 1110 0010	11 1110 0016	"	E388	"	TLE80 = (E388 - 994* S - E78)/S	"	"	"
	TLE81	453	"	"	"	11 1110 0011	11 1110 0666	"	E389	"	TLE81 = (E389 - 995* S - E78)/S	"	"	"
	TLE82	454	"	"	"	11 1110 1101	11 1110 1166	"	E390	"	TLE82 = (E390 - 1005* S - E78)/S	"	"	"
	TLE83	455	"	"	"	11 1110 1110	11 1110 1116	"	E391	"	TLE83 = (E391 - 1006* S - E78)/S	"	"	"
	TLE84	456	"	"	"	11 1110 1111	11 1116 6666	"	E392	"	TLE84 = (E392 - 1007* S - E78)/S	"	"	"
	TLE85	457	"	"	"	11 1111 0000	11 1111 0006	"	E393	"	TLE85 = (E393 - 1008* S - E78)/S	"	"	"
	TLE86	458	"	"	"	11 1111 0001	11 1111 0066	"	E394	"	TLE86 = (E394 - 1009* S - E78)/S	"	"	"
	TLE87	459	"	"	"	11 1111 0010	11 1111 0016	"	E395	"	TLE87 = (E395 - 1010* S - E78)/S	"	"	"
	TLE88	460	"	"	"	11 1111 0011	11 1111 0666	"	E396	"	TLE88 = (E396 - 1011* S - E78)/S	"	"	"
	TLE89	461	"	"	"	11 1111 0101	11 1111 0166	"	E397	"	TLE89 = (E397 - 1013* S - E78)/S	"	"	"
	TLE90	462	"	"	"	11 1111 0110	11 1111 0116	"	E398	"	TLE90 = (E398 - 1014* S - E78)/S	"	"	"
	TLE91	463	"	"	"	11 1111 0111	11 1111 0666	"	E399	"	TLE91 = (E399 - 1015* S - E78)/S	"	"	"
	TLE92	464	"	"	"	11 1111 1000	11 1111 1006	"	E400	"	TLE92 = (E400 - 1016* S - E78)/S	"	"	"
	TLE93	465	"	"	"	11 1111 1001	11 1111 1066	"	E401	"	TLE93 = (E401 - 1017* S - E78)/S	"	"	"
	TLE94	466	"	"	"	11 1111 1010	11 1111 1016	"	E402	"	TLE94 = (E402 - 1018* S - E78)/S	"	"	"
	TLE95	467	"	"	"	11 1111 1011	11 1111 1666	"	E403	"	TLE95 = (E403 - 1019* S - E78)/S	"	"	"
	TLE96	468	"	"	"	11 1111 1100	11 1111 1106	"	E404	"	TLE96 = (E404 - 1020* S - E78)/S	"	"	"
	TLE97	469	"	"	"	11 1111 1101	11 1111 1166	"	E405	"	TLE97 = (E405 - 1021* S - E78)/S	"	"	"
	TLE98	470	"	"	"	11 1111 1110	11 1111 1116	"	E406	"	TLE98 = (E406 - 1022* S - E78)/S	"	"	"

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltages ref. pin no. 14 (Figures 8, 9) 10 V_{CC} (V)	12 V_{EE} (V)	15 BpC (V)	Applied digital code-word 19<------>28 MSB<------>LSB	Output digital code-word 19<----->1,18,17> MSB<------->LSB	Measurement sense lines ref. pin no. 14 Pin	Value	Unit	Equations	Limits Min	Limits Max	Units
5 T_A = 125°C											For differential linearity error, $D_{LE(N)}=(E(N+408)-E(N+407))/S$ where $S = (E_{407} - E_{408})/1022$ for N = 1 to 119			
			+5.00	-15.00	0	11 1111 1110	11 1111 111ϕ	13	E_{407}	V				
			"	"	"	00 0000 0000	00 0000 000ϕ	"	E_{408}	"				
	D_{LE1}	471	"	"	"	00 0000 0001	00 0000 00ϕϕ	"	E_{409}	"	$D_{LE1} = (E_{409} - E_{408})/S$	-0.90	+1.00	LSB
	D_{LE2}	472	"	"	"	00 0000 0010	00 0000 001ϕ	"	E_{410}	"	$D_{LE2} = (E_{410} - E_{409})/S$	"	"	"
	D_{LE3}	473	"	"	"	00 0000 0011	00 0000 0ϕϕϕ	"	E_{411}	"	$D_{LE3} = (E_{411} - E_{410})/S$	"	"	"
	D_{LE4}	474	"	"	"	00 0000 0100	00 0000 010ϕ	"	E_{412}	"	$D_{LE4} = (E_{412} - E_{411})/S$	"	"	"
	D_{LE5}	475	"	"	"	00 0000 0101	00 0000 01ϕϕ	"	E_{413}	"	$D_{LE5} = (E_{413} - E_{412})/S$	"	"	"
	D_{LE6}	476	"	"	"	00 0000 0110	00 0000 011ϕ	"	E_{414}	"	$D_{LE6} = (E_{414} - E_{413})/S$	"	"	"
	D_{LE7}	477	"	"	"	00 0000 0111	00 0000 ϕϕϕϕ	"	E_{415}	"	$D_{LE7} = (E_{415} - E_{414})/S$	"	"	"
	D_{LE8}	478	"	"	"	00 0000 1000	00 0000 100ϕ	"	E_{416}	"	$D_{LE8} = (E_{416} - E_{415})/S$	"	"	"
	D_{LE9}	479	"	"	"	00 0000 1001	00 0000 10ϕϕ	"	E_{417}	"	$D_{LE9} = (E_{417} - E_{416})/S$	"	"	"
	D_{LE10}	480	"	"	"	00 0000 1010	00 0000 101ϕ	"	E_{418}	"	$D_{LE10} = (E_{418} - E_{417})/S$	"	"	"
	D_{LE11}	481	"	"	"	00 0000 1011	00 0000 1ϕϕϕ	"	E_{419}	"	$D_{LE11} = (E_{419} - E_{418})/S$	"	"	"
	D_{LE12}	482	"	"	"	00 0000 1100	00 0000 110ϕ	"	E_{420}	"		"	"	"
	D_{LE13}	483	"	"	"	00 0000 1101	00 0000 11ϕϕ	"	E_{421}	"	$D_{LE13} = (E_{421} - E_{420})/S$	"	"	"
	D_{LE14}	484	"	"	"	00 0000 1110	00 0000 111ϕ	"	E_{422}	"	$D_{LE14} = (E_{422} - E_{421})/S$	"	"	"
	D_{LE15}	485	"	"	"	00 0000 1111	00 000ϕ ϕϕϕϕ	"	E_{423}	"	$D_{LE15} = (E_{423} - E_{422})/S$	"	"	"
	D_{LE16}	486	"	"	"	00 0001 0000	00 0001 000ϕ	"	E_{424}	"	$D_{LE16} = (E_{424} - E_{423})/S$	"	"	"
	D_{LE17}	487	"	"	"	00 0001 0001	00 0001 00ϕϕ	"	E_{425}	"	$D_{LE17} = (E_{425} - E_{424})/S$	"	"	"
	D_{LE18}	488	"	"	"	00 0001 0010	00 0001 001ϕ	"	E_{426}	"	$D_{LE18} = (E_{426} - E_{425})/S$	"	"	"
	D_{LE19}	489	"	"	"	00 0001 0011	00 0001 0ϕϕϕ	"	E_{427}	"	$D_{LE19} = (E_{427} - E_{426})/S$	"	"	"
	D_{LE20}	490	"	"	"	00 0001 1100	00 0001 110ϕ	"	E_{428}	"		"	"	"
	D_{LE21}	491	"	"	"	00 0001 1101	00 0001 11ϕϕ	"	E_{429}	"	$D_{LE21} = (E_{429} - E_{428})/S$	"	"	"
	D_{LE22}	492	"	"	"	00 0001 1110	00 0001 111ϕ	"	E_{430}	"	$D_{LE22} = (E_{430} - E_{429})/S$	"	"	"
	D_{LE23}	493	"	"	"	00 0001 1111	00 000ϕ ϕϕϕϕ	"	E_{431}	"	$D_{LE23} = (E_{431} - E_{430})/S$	"	"	"
	D_{LE24}	494	"	"	"	00 0010 0000	00 0010 000ϕ	"	E_{432}	"	$D_{LE24} = (E_{432} - E_{431})/S$	"	"	"
	D_{LE25}	495	"	"	"	00 0010 0001	00 0010 00ϕϕ	"	E_{433}	"	$D_{LE25} = (E_{433} - E_{432})/S$	"	"	"
	D_{LE26}	496	"	"	"	00 0010 0010	00 0010 001ϕ	"	E_{434}	"	$D_{LE26} = (E_{434} - E_{433})/S$	"	"	"
	D_{LE27}	497	"	"	"	00 0010 0011	00 0010 0ϕϕϕ	"	E_{435}	"	$D_{LE27} = (E_{435} - E_{434})/S$	"	"	"
	D_{LE28}	498	"	"	"	00 0011 1100	00 0011 110ϕ	"	E_{436}	"		"	"	"
	D_{LE29}	499	"	"	"	00 0011 1101	00 0011 11ϕϕ	"	E_{437}	"	$D_{LE29} = (E_{437} - E_{436})/S$	"	"	"
	D_{LE30}	500	"	"	"	00 0011 1110	00 0011 111ϕ	"	E_{438}	"	$D_{LE30} = (E_{438} - E_{437})/S$	"	"	"
	D_{LE31}	501	"	"	"	00 0011 1111	00 0ϕϕϕ ϕϕϕϕ	"	E_{439}	"	$D_{LE31} = (E_{439} - E_{438})/S$	"	"	"
	D_{LE32}	502	"	"	"	00 0100 0000	00 0100 000ϕ	"	E_{440}	"	$D_{LE32} = (E_{440} - E_{439})/S$	"	"	"
	D_{LE33}	503	"	"	"	00 0100 0001	00 0100 00ϕϕ	"	E_{441}	"	$D_{LE33} = (E_{441} - E_{440})/S$	"	"	"
	D_{LE34}	504	"	"	"	00 0100 0010	00 0100 001ϕ	"	E_{442}	"	$D_{LE34} = (E_{442} - E_{441})/S$	"	"	"
	D_{LE35}	505	"	"	"	00 0100 0011	00 0100 0ϕϕϕ	"	E_{443}	"	$D_{LE35} = (E_{443} - E_{442})/S$	"	"	"
	D_{LE36}	506	"	"	"	00 0111 1100	00 0111 110ϕ	"	E_{444}	"		"	"	"
	D_{LE37}	507	"	"	"	00 0111 1101	00 0111 11ϕϕ	"	E_{445}	"	$D_{LE37} = (E_{445} - E_{444})/S$	"	"	"
	D_{LE38}	508	"	"	"	00 0111 1110	00 0111 111ϕ	"	E_{446}	"	$D_{LE38} = (E_{446} - E_{445})/S$	"	"	"
	D_{LE39}	509	"	"	"	00 0111 1111	00 ϕϕϕϕ ϕϕϕϕ	"	E_{447}	"	$D_{LE39} = (E_{447} - E_{446})/S$	"	"	"
	D_{LE40}	510	"	"	"	00 1000 0000	00 1000 000ϕ	"	E_{448}	"	$D_{LE40} = (E_{448} - E_{447})/S$	"	"	"
	D_{LE41}	511	"	"	"	00 1000 0001	00 1000 00ϕϕ	"	E_{449}	"	$D_{LE41} = (E_{449} - E_{448})/S$	"	"	"
	D_{LE42}	512	"	"	"	00 1000 0010	00 1000 001ϕ	"	E_{450}	"	$D_{LE42} = (E_{450} - E_{449})/S$	"	"	"
	D_{LE43}	513	"	"	"	00 1000 0011	00 1000 0ϕϕϕ	"	E_{451}	"	$D_{LE43} = (E_{451} - E_{450})/S$	"	"	"
	D_{LE44}	514	"	"	"	00 1111 1100	00 1111 110ϕ	"	E_{452}	"		"	"	"
	D_{LE45}	515	"	"	"	00 1111 1101	00 1111 11ϕϕ	"	E_{453}	"	$D_{LE45} = (E_{453} - E_{452})/S$	"	"	"

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltages ref. pin no. 14 (Figures 8, 9)			Applied digital code-word	Output digital code-word	Measurement sense lines ref. pin no. 14			Equations	Limits		Units
			10 V_{CC} (V)	12 V_{EE} (V)	15 SpC (V)	19<------>28 MSB<-----LSB	19<----->1,18,17 MSB<------LSB	Pin	Value	Unit		Min	Max	
5 T_A = 125°C	D_{LE46}	516	+5.00	-15.00	0	00 1111 1110	00 1111 111ϕ	13	E_{454}	V	$D_{LE46} = (E_{454} - E_{453})/S$	-0.90	+1.00	LSB
	D_{LE47}	517	"	"	"	00 1111 1111	00 ϕϕϕϕ ϕϕϕϕ	"	E_{455}	"	$D_{LE47} = (E_{455} - E_{454})/S$	"	"	"
	D_{LE48}	518	"	"	"	01 0000 0000	01 0000 000ϕ	"	E_{456}	"	$D_{LE48} = (E_{456} - E_{455})/S$	"	"	"
	D_{LE49}	519	"	"	"	01 0000 0001	01 0000 00ϕϕ	"	E_{457}	"	$D_{LE49} = (E_{457} - E_{456})/S$	"	"	"
	D_{LE50}	520	"	"	"	01 0000 0010	01 0000 001ϕ	"	E_{458}	"	$D_{LE50} = (E_{458} - E_{457})/S$	"	"	"
	D_{LE51}	521	"	"	"	01 0000 0011	01 0000 0ϕϕϕ	"	E_{459}	"	$D_{LE51} = (E_{459} - E_{458})/S$	"	"	"
	D_{LE52}	522	"	"	"	01 1111 1100	01 1111 110ϕ	"	E_{460}	"				
	D_{LE53}	523	"	"	"	01 1111 1101	01 1111 11ϕϕ	"	E_{461}	"	$D_{LE53} = (E_{461} - E_{460})/S$	"	"	"
	D_{LE54}	524	"	"	"	01 1111 1110	01 1111 111ϕ	"	E_{462}	"	$D_{LE54} = (E_{462} - E_{461})/S$	"	"	"
	D_{LE55}	525	"	"	"	01 1111 1111	ϕϕ ϕϕϕϕ ϕϕϕϕ	"	E_{463}	"	$D_{LE55} = (E_{463} - E_{462})/S$	"	"	"
	D_{LE56}	526	"	"	"	10 0000 0000	10 0000 000ϕ	"	E_{464}	"	$D_{LE56} = (E_{464} - E_{463})/S$	"	"	"
	D_{LE57}	527	"	"	"	10 0000 0001	10 0000 00ϕϕ	"	E_{465}	"	$D_{LE57} = (E_{465} - E_{464})/S$	"	"	"
	D_{LE58}	528	"	"	"	10 0000 0010	10 0000 001ϕ	"	E_{466}	"	$D_{LE58} = (E_{466} - E_{465})/S$	"	"	"
	D_{LE59}	529	"	"	"	10 0000 0011	10 0000 0ϕϕϕ	"	E_{467}	"	$D_{LE59} = (E_{467} - E_{466})/S$	"	"	"
	D_{LE60}	530	"	"	"	10 1111 1100	10 1111 110ϕ	"	E_{468}	"				
	D_{LE61}	531	"	"	"	10 1111 1101	10 1111 11ϕϕ	"	E_{469}	"	$D_{LE61} = (E_{469} - E_{468})/S$	"	"	"
	D_{LE62}	532	"	"	"	10 1111 1110	10 1111 111ϕ	"	E_{470}	"	$D_{LE62} = (E_{470} - E_{469})/S$	"	"	"
	D_{LE63}	533	"	"	"	10 1111 1111	1ϕ ϕϕϕϕ ϕϕϕϕ	"	E_{471}	"	$D_{LE63} = (E_{471} - E_{470})/S$	"	"	"
	D_{LE64}	534	"	"	"	11 0000 0000	11 0000 000ϕ	"	E_{472}	"	$D_{LE64} = (E_{472} - E_{471})/S$	"	"	"
	D_{LE65}	535	"	"	"	11 0000 0001	11 0000 00ϕϕ	"	E_{473}	"	$D_{LE65} = (E_{473} - E_{472})/S$	"	"	"
	D_{LE66}	536	"	"	"	11 0000 0010	11 0000 001ϕ	"	E_{474}	"	$D_{LE66} = (E_{474} - E_{473})/S$	"	"	"
	D_{LE67}	537	"	"	"	11 0000 0011	11 0000 0ϕϕϕ	"	E_{475}	"	$D_{LE67} = (E_{475} - E_{474})/S$	"	"	"
	D_{LE68}	538	"	"	"	11 0111 1100	11 0111 110ϕ	"	E_{476}	"				
	D_{LE69}	539	"	"	"	11 0111 1101	11 0111 11ϕϕ	"	E_{477}	"	$D_{LE69} = (E_{477} - E_{476})/S$	"	"	"
	D_{LE70}	540	"	"	"	11 0111 1110	11 0111 111ϕ	"	E_{478}	"	$D_{LE70} = (E_{478} - E_{477})/S$	"	"	"
	D_{LE71}	541	"	"	"	11 0111 1111	11 ϕϕϕϕ ϕϕϕϕ	"	E_{479}	"	$D_{LE71} = (E_{479} - E_{478})/S$	"	"	"
	D_{LE72}	542	"	"	"	11 1000 0000	11 1000 000ϕ	"	E_{480}	"	$D_{LE72} = (E_{480} - E_{479})/S$	"	"	"
	D_{LE73}	543	"	"	"	11 1000 0001	11 1000 00ϕϕ	"	E_{481}	"	$D_{LE73} = (E_{481} - E_{480})/S$	"	"	"
	D_{LE74}	544	"	"	"	11 1000 0010	11 1000 001ϕ	"	E_{482}	"	$D_{LE74} = (E_{482} - E_{481})/S$	"	"	"
	D_{LE75}	545	"	"	"	11 1000 0011	11 1000 0ϕϕϕ	"	E_{483}	"	$D_{LE75} = (E_{483} - E_{482})/S$	"	"	"
	D_{LE76}	546	"	"	"	11 1011 1100	11 1011 110ϕ	"	E_{484}	"				
	D_{LE77}	547	"	"	"	11 1011 1101	11 1011 11ϕϕ	"	E_{485}	"	$D_{LE77} = (E_{485} - E_{484})/S$	"	"	"
	D_{LE78}	548	"	"	"	11 1011 1110	11 1011 111ϕ	"	E_{486}	"	$D_{LE78} = (E_{486} - E_{485})/S$	"	"	"
	D_{LE79}	549	"	"	"	11 1011 1111	11 1ϕϕϕ ϕϕϕϕ	"	E_{487}	"	$D_{LE79} = (E_{487} - E_{486})/S$	"	"	"
	D_{LE80}	550	"	"	"	11 1100 0000	11 1100 000ϕ	"	E_{488}	"	$D_{LE80} = (E_{488} - E_{487})/S$	"	"	"
	D_{LE81}	551	"	"	"	11 1100 0001	11 1100 00ϕϕ	"	E_{489}	"	$D_{LE81} = (E_{489} - E_{488})/S$	"	"	"
	D_{LE82}	552	"	"	"	11 1100 0010	11 1100 001ϕ	"	E_{490}	"	$D_{LE82} = (E_{490} - E_{489})/S$	"	"	"
	D_{LE83}	553	"	"	"	11 1100 0011	11 1100 0ϕϕϕ	"	E_{491}	"	$D_{LE83} = (E_{491} - E_{490})/S$	"	"	"
	D_{LE84}	554	"	"	"	11 1101 1100	11 1101 110ϕ	"	E_{492}	"				
	D_{LE85}	555	"	"	"	11 1101 1101	11 1101 11ϕϕ	"	E_{493}	"	$D_{LE85} = (E_{493} - E_{492})/S$	"	"	"
	D_{LE86}	556	"	"	"	11 1101 1110	11 1101 111ϕ	"	E_{494}	"	$D_{LE86} = (E_{494} - E_{493})/S$	"	"	"
	D_{LE87}	557	"	"	"	11 1101 1111	11 11ϕϕ ϕϕϕϕ	"	E_{495}	"	$D_{LE87} = (E_{495} - E_{494})/S$	"	"	"
	D_{LE88}	558	"	"	"	11 1110 0000	11 1110 000ϕ	"	E_{496}	"	$D_{LE88} = (E_{496} - E_{495})/S$	"	"	"
	D_{LE89}	559	"	"	"	11 1110 0001	11 1110 00ϕϕ	"	E_{497}	"	$D_{LE89} = (E_{497} - E_{496})/S$	"	"	"
	D_{LE90}	560	"	"	"	11 1110 0010	11 1110 001ϕ	"	E_{498}	"	$D_{LE90} = (E_{498} - E_{497})/S$	"	"	"
	D_{LE91}	561	"	"	"	11 1110 0011	11 1110 0ϕϕϕ	"	E_{499}	"	$D_{LE91} = (E_{499} - E_{498})/S$	"	"	"
	D_{LE92}	562	"	"	"	11 1110 1100	11 1110 110ϕ	"	E_{500}	"				
	D_{LE93}	563	"	"	"	11 1110 1101	11 1110 11ϕϕ	"	E_{501}	"	$D_{LE93} = (E_{501} - E_{500})/S$	"	"	"
	D_{LE94}	564	"	"	"	11 1110 1110	11 1110 111ϕ	"	E_{502}	"	$D_{LE94} = (E_{502} - E_{501})/S$	"	"	"
	D_{LE95}	565	"	"	"	11 1110 1111	11 111ϕ ϕϕϕϕ	"	E_{503}	"	$D_{LE95} = (E_{503} - E_{502})/S$	"	"	"
	D_{LE96}	566	"	"	"	11 1111 0000	11 1111 000ϕ	"	E_{504}	"	$D_{LE96} = (E_{504} - E_{503})/S$	"	"	"
	D_{LE97}	567	"	"	"	11 1111 0001	11 1111 00ϕϕ	"	E_{505}	"	$D_{LE97} = (E_{505} - E_{504})/S$	"	"	"

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TABLE III. Group A inspection Continued.

Subgroup	Symbol	Test no.	Applied voltages ref. pin no. 14 (Figures 8, 9) 10 V_{CC} (V)	12 V_{EE} (V)	15 βpC (V)	Applied digital code word 19<------>2 MSB<------>LSB	Output digital code word 19<------>1,18,17 MSB<------>LSB	Measurement sense lines ref. pin no. 14 Pin	Value	Unit	Equations	Limits Min	Max	Units
5 T_A = 125°C	D_{LE98}	568	+5.00	-15.00	0	11 1111 0010	11 1111 001ø	13	E_{506}	V	$D_{LE98} = (E_{506} - E_{505})/S$	-0.90	+1.00	LSB
	D_{LE99}	569	"	"	"	11 1111 0011	11 1111 0ööö	"	E_{507}	"	$D_{LE99} = (E_{507} - E_{506})/S$	"	"	"
	D_{LE100}	570	"	"	"	11 1111 0100	11 1111 010ö	"	E_{508}	"		"	"	"
	D_{LE101}	571	"	"	"	11 1111 0101	11 1111 01öö	"	E_{509}	"	$D_{LE101} = (E_{509} - E_{508})/S$	"	"	"
	D_{LE102}	572	"	"	"	11 1111 0110	11 1111 011ø	"	E_{510}	"	$D_{LE102} = (E_{510} - E_{509})/S$	"	"	"
	D_{LE103}	573	"	"	"	11 1111 0111	11 1111 0ööö	"	E_{511}	"	$D_{LE103} = (E_{511} - E_{510})/S$	"	"	"
	D_{LE104}	574	"	"	"	11 1111 1000	11 1111 100ø	"	E_{512}	"	$D_{LE104} = (E_{512} - E_{511})/S$	"	"	"
	D_{LE105}	575	"	"	"	11 1111 1001	11 1111 10öö	"	E_{513}	"	$D_{LE105} = (E_{513} - E_{512})/S$	"	"	"
	D_{LE106}	576	"	"	"	11 1111 1010	11 1111 101ø	"	E_{514}	"	$D_{LE106} = (E_{514} - E_{513})/S$	"	"	"
	D_{LE107}	577	"	"	"	11 1111 1011	11 1111 1ööö	"	E_{515}	"	$D_{LE107} = (E_{515} - E_{514})/S$	"	"	"
	D_{LE108}	578	"	"	"	11 1111 1100	11 1111 110ø	"	E_{516}	"	$D_{LE108} = (E_{516} - E_{515})/S$	"	"	"
	D_{LE109}	579	"	"	"	11 1111 1101	11 1111 11öö	"	E_{517}	"	$D_{LE109} = (E_{517} - E_{516})/S$	"	"	"
	D_{LE110}	580	"	"	"	11 1111 1110	11 1111 111ø	"	E_{518}	"	$D_{LE110} = (E_{518} - E_{517})/S$	"	"	"
			+5.00	-15.00	"	11 1111 1110	11 1111 111ø	"	E_{519}	"	For S = $(E_{519} - E_{520})/1022$			
			"	"	"	00 0000 0000	00 0000 000ø	"	E_{520}	"				
	$-P_{SSa}$	581	"	-13.50	"	" " "	" " "	"	E_{521}	"	$-P_{SSa} = (E_{521} - E_{520})/(10.00*S)$	-0.18	0.18	LSB/%PS
			"	-16.00	"	" " "	" " "	"	E_{522}	"	$-P_{SSa} = (E_{520} - E_{522})/(6.67*S)$	"	"	"
	$-P_{SSb}$	582	"	-13.50	"	11 1111 1110	11 1111 111ø	"	E_{523}	"	$-P_{SSb} = (E_{523} - E_{519})/(10.00*S)$	"	"	"
			"	-16.00	"	" " "	" " "	"	E_{524}	"	$-P_{SSb} = (E_{519} - E_{524})/(6.67*S)$	"	"	"
			+5.00	-15.00	"	11 1111 1110	11 1111 111ø	"	E_{525}	"	For S = $(E_{525} - E_{526})/1022$			
			"	"	"	00 0000 0000	00 0000 000ø	"	E_{526}	"				
	$+P_{SSa}$	583	+5.50	"	"	" " "	" " "	"	E_{527}	"	$+P_{SSa} = (E_{527} - E_{526})/(10.00*S)$	-0.15	0.15	"
			+4.50	"	"	" " "	" " "	"	E_{528}	"	$+P_{SSa} = (E_{526} - E_{528})/(10.00*S)$	"	"	"
	$+P_{SSb}$	584	+5.50	"	"	11 1111 1110	11 1111 111ø	"	E_{529}	"	$+P_{SSb} = (E_{529} - E_{525})/(10.00*S)$	"	"	"
			+4.50	"	"	" " "	" " "	"	E_{530}	"	$+P_{SSb} = (E_{525} - E_{530})/(10.00*S)$	"	"	"
6 T_A = -55°C	(All test parameters, test conditions, equations, and test limits are identical with those specified in table III, subgroup 5.)													
7 T_A = 25°C		802 803	+5.00	-15.00	0	00 0000 0000 11 1111 1110	00 0000 000ø 11 1111 111ø	13	E_{760} E_{761}	V	For end-point linearity error, $TLE(N) = (E(N*761) - V(ideal))/S$ where S = $(E_{761} - E_{760})/1022$ for N = 1 to 1022			
	T_{LEmp}	804	Determine the code-word and the value of the maximum positive digital input code-words from (00 0000 0000) to (11 1111 1111). +5.00	-15.00	0	<----N1---->	<----N1---->	13	E_{762}	V	Linearity error for all combinations of the $T_{LEmp} = 2.05*(E_{762} - N1*S - E_{760})$	-.50	+.50	LSB
	T_{LEmn}	805	Determine the code-word and the value of the maximum negative digital input code-words from (00 0000 0000) to (11 1111 1111). +5.00	-15.00	0	<----N1---->	<----N1---->	13	E_{763}	V	Linearity error for all combinations of the $T_{LEmn} = 2.05*(E_{763} - N1*S - E_{760})$	-.50	+.50	LSB
For transition linearity error, the worst positive and negative error values, as determined by the manufacturer's abbreviated transition linearity (TLE) test procedure for subgroup 1 shall be within 150 millLSB of the worst positive and negative error values, as determined by the all codes test.														

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TABLE III. Group A inspection - Continued.

Subgroups	Symbol	Test no.	Applied voltages ref. pin no. 14 (Figures 8, 9)			Applied digital code word	Output digital code word	Measurement sense lines ref. pin no. 14			Equations	Limits		Units
			10 V_{CC} (V)	12 V_{EE} (V)	15 V_{DD} (V)	[illegible] MSB [illegible] LSB	[illegible] MSB [illegible] LSB	Pin	Value	Unit		Min	Max	
8 T_A = 125°C	(All test parameters, test conditions, equations, and test limits are identical with those specified in table III, subgroup 7.)													
	For transition linearity error, the worst positive and negative error values, as determined by the manufacturer's abbreviated transition linearity (TLE) test procedure for subgroup 2 shall be within 150 milliLSB of the worst positive and negative error values, as determined by the all codes test.													
8 T_A = -55°C	(All test parameters, test conditions, equations, and test limits are identical with those specified in table III, subgroup 7.)													
	For transition linearity error, the worst positive and negative error values, as determined by the manufacturer's abbreviated transition linearity (TLE) test procedure for subgroup 3 shall be within 150 milliLSB of the worst positive and negative error values, as determined by the all codes test.													

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	Applied voltages and loads ref. pin no. 14 (Figure 7)						Measurement sense lines ref. pin no. 14			Equations	Limits		Units
			10 V_{CC} (V)	12 V_{EE} (V)	15 B & C (V)	13 Vin(a) (V)	Output Pin	Output Load	Pin	Value	Unit		Min	Max	
9 T_A = 25°C	t_{D01}	814	+5.00	-15.00	-	MAX(-)	9	1.6 mA	9	t_2	ns	$t_{D01} = t_2$ (3-STATE --> LOW)	-	500	ns
	t_{D02}	815	"	"	"	"	8	"	8	t_3	"	$t_{D02} = t_3$	"	"	"
	t_{D03}	816	"	"	"	"	7	"	7	t_4	"	$t_{D03} = t_4$	"	"	"
	t_{D04}	817	"	"	"	"	6	"	6	t_5	"	$t_{D04} = t_5$	"	"	"
	t_{D05}	818	"	"	"	"	5	"	5	t_6	"	$t_{D05} = t_6$	"	"	"
	t_{D06}	819	"	"	"	"	4	"	4	t_7	"	$t_{D06} = t_7$	"	"	"
	t_{D07}	820	"	"	"	"	3	"	3	t_8	"	$t_{D07} = t_8$	"	"	"
	t_{D08}	821	"	"	"	"	2	"	2	t_9	"	$t_{D08} = t_9$	"	"	"
	t_{D09}	822	"	"	"	"	1	"	1	t_{10}	"	$t_{D09} = t_{10}$	"	"	"
	t_{D010}	823	"	"	"	"	18	"	18	t_{11}	"	$t_{D010} = t_{11}$	"	"	"
	t_{D011}	824	"	"	"	MAX(+)	9	-1.0 mA	9	t_{12}	"	$t_{D011} = t_{12}$ (3-STATE --> HIGH)	-	-	"
	t_{D012}	825	"	"	"	"	8	"	8	t_{13}	"	$t_{D012} = t_{13}$	"	"	"
	t_{D013}	826	"	"	"	"	7	"	7	t_{14}	"	$t_{D013} = t_{14}$	"	"	"
	t_{D014}	827	"	"	"	"	6	"	6	t_{15}	"	$t_{D014} = t_{15}$	"	"	"
	t_{D015}	828	"	"	"	"	5	"	5	t_{16}	"	$t_{D015} = t_{16}$	"	"	"
	t_{D016}	829	"	"	"	"	4	"	4	t_{17}	"	$t_{D016} = t_{17}$	"	"	"
	t_{D017}	830	"	"	"	"	3	"	3	t_{18}	"	$t_{D017} = t_{18}$	"	"	"
	t_{D018}	831	"	"	"	"	2	"	2	t_{19}	"	$t_{D018} = t_{19}$	"	"	"
	t_{D019}	832	"	"	"	"	1	"	1	t_{20}	"	$t_{D019} = t_{20}$	"	"	"
	t_{D020}	833	"	"	"	"	18	"	18	t_{21}	"	$t_{D020} = t_{21}$	"	"	"
	t_{bk1}	835	"	"	-	MAX(-)	9	1.6 mA	9	t_{22}	µs	$t_{DR1} = t_{22}$ (LOW --> 3 STATE)	0.1	10.0	µs
	t_{bk2}	836	"	"	"	"	8	"	8	t_{23}	"	$t_{DR2} = t_{23}$	"	"	"
	t_{bk3}	837	"	"	"	"	7	"	7	t_{24}	"	$t_{DR3} = t_{24}$	"	"	"
	t_{bk4}	838	"	"	"	"	6	"	6	t_{25}	"	$t_{DR4} = t_{25}$	"	"	"
	t_{bk5}	839	"	"	"	"	5	"	5	t_{26}	"	$t_{DR5} = t_{26}$	"	"	"
	t_{bk6}	840	"	"	"	"	4	"	4	t_{27}	"	$t_{DR6} = t_{27}$	"	"	"
	t_{bk7}	841	"	"	"	"	3	"	3	t_{28}	"	$t_{DR7} = t_{28}$	"	"	"
	t_{bk8}	842	"	"	"	"	2	"	2	t_{29}	"	$t_{DR8} = t_{29}$	"	"	"
	t_{bk9}	843	"	"	"	"	1	"	1	t_{30}	"	$t_{DR9} = t_{30}$	"	"	"
	t_{bk10}	844	"	"	"	"	18	"	18	t_{31}	"	$t_{DR10} = t_{31}$	"	"	"
	t_{bk11}	845	"	"	"	MAX(+)	9	-1.0 mA	9	t_{32}	µs	$t_{DR11} = t_{32}$ (HIGH --> 3 STATE)	"	"	"
	t_{bk12}	846	"	"	"	"	8	"	8	t_{33}	"	$t_{DR12} = t_{33}$	"	"	"
	t_{bk13}	847	"	"	"	"	7	"	7	t_{34}	"	$t_{DR13} = t_{34}$	"	"	"
	t_{bk14}	848	"	"	"	"	6	"	6	t_{35}	"	$t_{DR14} = t_{35}$	"	"	"
	t_{bk15}	849	"	"	"	"	5	"	5	t_{36}	"	$t_{DR15} = t_{36}$	"	"	"
	t_{bk16}	850	"	"	"	"	4	"	4	t_{37}	"	$t_{DR16} = t_{37}$	"	"	"
	t_{bk17}	851	"	"	"	"	3	"	3	t_{38}	"	$t_{DR17} = t_{39}$	"	"	"
	t_{bk18}	852	"	"	"	"	2	"	2	t_{39}	"	$t_{DR18} = t_{39}$	"	"	"
	t_{bk19}	853	"	"	"	"	1	"	1	t_{40}	"	$t_{DR19} = t_{40}$	"	"	"
	t_{bk20}	854	"	"	"	"	18	"	18	t_{41}	"	$t_{DR20} = t_{41}$	"	"	"
	RI	855	"	"	"0"	1/2 VFS	-	-	13	I_{82}	mA	RI = Vin(a)/I_{82}	3	7	kohms
	$t_{\overline{DR}}$	856	"	"	-	-			17	t_{42}	us	Refer figure 4	0.1	1.5	us

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 10, 11 and 12 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Electrical parameters shall be as specified in table II herein. For class S devices, delta limits shall apply only to subgroup 5 of group B inspection.

b. Steady state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using the circuit shown on figure 6.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B devices.

b. Steady state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition A, using the circuit shown on figure 6.

(2) T_A = 125°C minimum.

(3) Test duration: 1000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. A special subgroup shall be added to the group C inspection requirements and shall consist of the tests, conditions, and limits specified for subgroups 7 and 8 of group A. The LTPD for the special subgroup shall be 15 for all classes.

TABLE IV. Group C delta limits at T_A = 25°C.

Table III test no.	Symbol	Delta		EOL limits		Units
		Min	Max	Min	Max	
151	ZE	-1.5	1.5	-2.5	2.5	LSB
152	BZE	-2.0	2.0	-3.0	3.0	"
153	$-V_{FSE}$	-4.5	4.5	-8.2	8.2	"
154	$+V_{FSE}$	"	"	"	"	"
155	$+V_{FSE}$	"	"	"	"	"

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cooldown procedures. When the devices are measured at 25°C following application of the steady state life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias.

4.6 Data reporting. When specified in the acquisition document, a copy of the following data, as applicable, shall be supplied:

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and operating life tests.

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parameter data (see 4.2b).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing design application equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable, these requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

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6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331.

BpC	-	Bipolar control
B & $\overline{C}$	-	Blank and convert
B_{ZE}	-	Bipolar zero error
DLE	-	Differential linearity error
$\overline{DR}$	-	Data ready
I_{EE}, I_{CC}	-	Input currents for negative and logic voltage supplies, respectively.
I_{IH}	-	Digital high level input current
I_{IL}	-	Digital low level input current
I_{ZH}	-	Digital output impedance (output forced high)
I_{ZL}	-	Digital output impedance (output forced low)
LSB	-	Least significant bit
MSB	-	Most significant bit
P_{SS}	-	Power supply sensitivity
Ri	-	Input resistance of the analog input
t_C	-	Conversion time
t_{DO}	-	Output data ready delay time after high to low $\overline{DR}$ transition
t_{DR}	-	Output data blanking delay time after low to high B & $\overline{C}$ transition
T_{LE}	-	Integral transition linearity error
V_{FSE}	-	Full scale voltage error
Vin (a)	-	Analog input voltage
V_{OH}	-	Digital high level output voltage
V_{OL}	-	Digital low level output voltage transition
Z_E	-	Input offset voltage for zero output code transition (Zero error)
MAX (+)	-	The positive voltage required at the analog voltage input pin to produce a digital code of 11 1111 1111 at the DUT output pins.
MAX (-)	-	The negative voltage required at the analog voltage input pin to produce a digital code of 00 0000 0000 at the DUT output pins.

Bipolar Control (BpC): The bipolar control signal is an input signal that controls the operating mode of the converter. When the control pin is connected to digital ground, the converter operates in the unipolar mode. When the control pin is left open, the converter operates in the bipolar mode.

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Bipolar Mode: Bipolar mode is the A/D converter operation mode that includes both positive and negative input analog voltages and provides an offset binary output code.

Blank and $\overline{\text{Convert}}$ (B & $\overline{\text{C}}$): The blank and $\overline{\text{convert}}$ signal is the input signal that controls the conversion mode of the converter. When the B & $\overline{\text{C}}$ signal is brought low, the $\overline{\text{DR}}$ signal is high, the data output pins are open and the converter is in the conversion mode. When the B & $\overline{\text{C}}$ signal is low and the $\overline{\text{DR}}$ signal is low, the conversion is complete and the data output pins are active with new data. When the B & $\overline{\text{C}}$ signal is brought high, the $\overline{\text{DR}}$ signal goes high, the data output pins are open and the converter is readied for a new conversion.

Data Ready ($\overline{\text{DR}}$): $\overline{\text{Data ready}}$ is a digital output signal that indicates the conversion status of the converter. When this signal is a logic "0", the conversion is complete.

Differential Linearity: Differential linearity is the difference between two input analog voltages required to establish adjacent output code-word transitions. The ideal differential linearity is 1 LSB.

Differential Linearity Error: Differential linearity error is the difference between the actual and the ideal differential linearity values for any two adjacent code-word transitions.

Full-scale Range: Full-scale range is the voltage difference between the input voltages at the first bit transition and the last bit transition plus twice the voltage difference between two adjacent bit transitions.

Integral Transition Linearity Error: Integral transition linearity error is the difference between the measured analog least significant bit is the average difference between the respect to the ideal voltage at the same integral transition points as defined by a straight line that passes through the first and the last integral transition points.

Least Significant Bit: The least significant bit is the bit in the output code that carries the least weight. The value of the analog input voltages at two adjacent output bit transitions. The ideal difference voltage is the value of the full scale range divided by 1024.

Monotonicity: A device is monotonic if the ratio of the incremental change in the output to an incremental change in the input (O/I) does not change sign over the entire operating range.

Most Significant Bit: The most significant bit is the bit in the output code that carries the most weight. It is the output bit that changes state when the analog input voltage change from its most positive value to one-half of the full scale range.

6.4 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	571

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Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - AR, SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/134	Microcircuits, Linear, Analog-To-Digital Convert With 3-State Buffered 10 BIT Outputs, Monolithic Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify):
b. ADDRESS (Street, City, State, ZIP Code)	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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